

# Pseudospectral connectedness thresholds and strict dimension monotonicity for asymmetric tridiagonal Toeplitz matrices

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## Abstract

For the asymmetric tridiagonal Toeplitz matrix  $A_n(a) = \text{tridiag}(a, 0, 1)$ ,  $0 < a < 1$ , we determine the connectedness threshold of its strict Euclidean  $\varepsilon$ -pseudospectrum. Define

$$\gamma_n = \max_{|x| \leq 2\sqrt{a} \cos(\pi/(n+1))} s_{\min}(xI_n - A_n).$$

We prove that the least singular value of  $(x + iy)I_n - A_n$  is nondecreasing in  $|y|$ . Hence every pseudospectral component is contractible and

$$\sigma_\varepsilon(A_n(a)) \text{ is connected} \iff \varepsilon > \gamma_n.$$

Our main finite-size result is the strict dimension monotonicity  $\gamma_{n+1} < \gamma_n$  for every  $n \geq 2$ ; consequently, connectedness persists at every larger dimension. Explicit two-sided bounds give  $\gamma_n \asymp_a (n+1)a^{n/2}$  and

$$N_c(a, \varepsilon) = \frac{2}{\log(1/a)} \left( \log(1/\varepsilon) + \log \log(1/\varepsilon) \right) + O_a(1) \quad (\varepsilon \downarrow 0).$$

The proof combines a symmetric signed pencil, folded Chebyshev coordinates, interlacing, and determinant monotonicity.

*Keywords:* pseudospectrum, tridiagonal Toeplitz matrix, nonnormal matrix, least singular value, matrix perturbation, Chebyshev polynomial

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## 1. Introduction

For a matrix  $A \in \mathbb{C}^{n \times n}$  and  $\varepsilon > 0$ , the strict Euclidean  $\varepsilon$ -pseudospectrum is

$$\sigma_\varepsilon(A) = \sigma(A) \cup \{z \in \mathbb{C} : \|(zI_n - A)^{-1}\|_2 > \varepsilon^{-1}\} = \{z \in \mathbb{C} : s_{\min}(zI_n - A) < \varepsilon\}. \quad (1.1)$$

Equivalently, it is the union of the spectra of  $A + \Delta A$  over all complex perturbations satisfying  $\|\Delta A\|_2 < \varepsilon$  [13]. Pseudospectral components therefore encode robust spectral clusters under an unstructured norm bound. Their merger is also tied to distance-to-defectivity questions: first coalescence is characterized by a generalized saddle point of the smallest-singular-value function, and its level determines the spectral-norm distance to defectivity [1]. These observations motivate a finite-dimensional topological problem that is not determined by the spectrum alone.

We solve this problem for the asymmetric tridiagonal Toeplitz family

$$A_n(a) = \text{tridiag}(a, 0, 1) = \begin{bmatrix} 0 & 1 & & \\ a & 0 & \ddots & \\ & \ddots & \ddots & 1 \\ & & a & 0 \end{bmatrix}, \quad n \geq 2, \quad 0 < a < 1, \quad (1.2)$$

where the arguments of  $\text{tridiag}$  are the subdiagonal, diagonal, and superdiagonal. Put  $r = \sqrt{a}$ , let  $V_n = \text{tridiag}(0, 0, 1)$ , and define

$$\mathcal{G}_n(\varrho) = \text{diag}(1, \varrho, \dots, \varrho^{n-1}).$$

Then

$$A_n(a) = \mathcal{G}_n(r) r(V_n + V_n^T) \mathcal{G}_n(r)^{-1}. \quad (1.3)$$

Thus  $A_n(a)$  is diagonally similar to a symmetric Jacobi matrix and has the simple real eigenvalues  $2r \cos(k\pi/(n+1))$ ,  $1 \leq k \leq n$ . The similarity is not unitary, however, so it does not preserve singular values or Euclidean pseudospectra. Indeed,  $\kappa_2(\mathcal{G}_n(r)) = r^{-(n-1)}$ , which already suggests an exponentially dimension-dependent pseudospectral scale.

Our question is global in the spectral parameter and discrete in the matrix size: for fixed  $(a, \varepsilon)$ , when is  $\sigma_\varepsilon(A_n(a))$  connected, and can connectedness be lost again when  $n$  is replaced by  $n+1$ ? At each size we measure the final obstruction by the largest real-axis backward-error barrier  $\gamma_n$ , obtained by maximizing  $s_{\min}(xI_n - A_n(a))$  over the spectral interval. The spectral endpoints and all interior eigenvalues are zeros, so this quantity is equivalently the largest maximum over the consecutive spectral gaps.

The results have four parts. First, vertical monotonicity of the least singular value reduces the planar topology to the real axis and implies that every pseudospectral component is contractible. Second, the exact criterion

$$\sigma_\varepsilon(A_n(a)) \text{ is connected} \iff \varepsilon > \gamma_n$$

reduces connectedness to one variational barrier. Third, we prove the finite-size inequality  $\gamma_{n+1} < \gamma_n$  for every  $n \geq 2$ , so connectedness persists with dimension. Finally, explicit two-sided estimates yield

$$\gamma_n \asymp_a (n+1)a^{n/2} \tag{1.4}$$

and an asymptotically sharp inversion of the connectedness threshold. Here and throughout, constants implicit in  $\asymp_a$  may depend on the fixed parameter  $a$ , but not on  $n$ .

The explicit bounds capture both the exponential factor and the sharp polynomial prefactor. The singular and normal endpoints  $a = 0, 1$  are discussed in section 7.

The relevant literature has several complementary strands. A foundational spectral limit-set theorem for banded Toeplitz matrices is due to Schmidt and Spitzer [12]. Finite Toeplitz spectra, inverse norms, and pseudospectra are studied in [5, 11], and tridiagonal Toeplitz structure is treated in [10]. For general matrices, optimization methods for pseudospectra appear in [6], while component coalescence and nearest defective matrices are treated in [1]. Recent work on tridiagonal  $k$ -Toeplitz matrices addresses broader periodic coefficient patterns [3, 4]. Our contribution is a finite-size order law for the global barrier  $\gamma_n$ , coupled with an exact planar connectedness criterion and sharp transition asymptotics.

The proof uses three reductions. A pentadiagonal determinant certificate first establishes vertical monotonicity. Persymmetry then turns singular values into signed eigenvalues of a symmetric pencil. Folded continuants and Chebyshev divided differences parametrize the relevant signed branch; chord comparisons control noncentral gaps, while interlacing and a separate scalar certification handle the parity-changing central gaps. Finally, rectangular factorizations give the sharp two-sided bounds and an elementary logarithmic inversion gives the threshold asymptotic. The longer algebraic certificates are isolated in appendices Appendix A to Appendix F.

## 2. Matrix family and main results

This section formulates connectedness as an exact comparison between the perturbation level and a real-axis barrier. The resulting scalar is a global

pointwise backward error: it measures the largest perturbation that must be crossed between consecutive eigenvalues.

Fix  $n \geq 2$ ,  $0 < a < 1$ , and write  $A_n = A_n(a)$ , with  $A_n(a)$  and  $\sigma_\varepsilon(A_n)$  as in (1.2) and (1.1). Define the first connected dimension by

$$N_c(a, \varepsilon) = \min\{n \geq 2 : \sigma_\varepsilon(A_n(a)) \text{ is connected}\}.$$

The theorem below proves that this set is nonempty and that every larger dimension is also connected.

Recall that  $r = \sqrt{a}$ , and note that  $A_n = V_n + aV_n^T$ .

Define

$$I_n^{\text{sp}} = \left[-2r \cos \frac{\pi}{n+1}, 2r \cos \frac{\pi}{n+1}\right], \quad g_n(x) = s_{\min}(xI_n - A_n),$$

$$\gamma_n = \max_{x \in I_n^{\text{sp}}} g_n(x).$$

The pointwise spectral backward-error identity is

$$g_n(x) = \min\{\|\Delta A\|_2 : \Delta A \in \mathbb{C}^{n \times n}, x \in \sigma(A_n + \Delta A)\}. \quad (2.1)$$

Thus  $\gamma_n$  is a global barrier, although the optimizing perturbation may depend on  $x$ . The explicit lower and upper barriers used below are

$$\mathcal{U}_n(a) = r^n \csc \frac{\pi}{n+1}, \quad (2.2)$$

$$\mathcal{L}_n(a) = \left(1 + r^2 - 2r \cos \frac{\pi}{n+1}\right) \frac{r^n(1 - r^2)}{(1 - r^{2(n+1)}) \sin \frac{3\pi}{2(n+1)}}. \quad (2.3)$$

**Theorem 2.1** (Main theorem). *For every  $0 < a < 1$  and  $\varepsilon > 0$ , the following assertions hold.*

1. *For every  $n \geq 2$ , every connected component of  $\sigma_\varepsilon(A_n(a))$  is contractible.*
2. *For  $n \geq 2$ , the barrier sequence is strictly decreasing,  $\gamma_{n+1} < \gamma_n$ , tends to zero, and*

$$\sigma_\varepsilon(A_n(a)) \text{ is connected} \iff \varepsilon > \gamma_n.$$

3. *Connectedness persists after its first occurrence:*

$$\{n \geq 2 : \sigma_\varepsilon(A_n(a)) \text{ is connected}\} = \{n \geq N_c(a, \varepsilon)\}.$$

Moreover,

$$1 + \max\left(\{1\} \cup \{n \geq 2 : \mathcal{L}_n(a) \geq \varepsilon\}\right) \leq N_c(a, \varepsilon),$$

$$N_c(a, \varepsilon) \leq \min\{n \geq 2 : \mathcal{U}_n(a) < \varepsilon\}. \quad (2.4)$$

4. For fixed  $a$ , as  $\varepsilon \downarrow 0$ ,

$$N_c(a, \varepsilon) = \frac{2}{\log(1/a)} (\log(1/\varepsilon) + \log \log(1/\varepsilon)) + O_a(1), \quad (2.5)$$

and  $N_c(a, \varepsilon) = 2$  if and only if  $\varepsilon > a$ .

The variational barrier  $\gamma_n$  is the exact connectedness threshold at dimension  $n$ . Its strict decrease gives the complete order law: all dimensions below  $N_c$  are disconnected and all dimensions at or above  $N_c$  are connected.

The normalization in (1.2) extends by affine reduction to tridiagonal Toeplitz matrices with positive unequal off-diagonals.

**Corollary 2.2** (Positive-off-diagonal Toeplitz matrices). *Let*

$$T_n = \text{tridiag}(\alpha, d, \beta), \quad d \in \mathbb{C}, \quad \alpha, \beta > 0, \quad \alpha \neq \beta,$$

where  $n \geq 2$ , and let  $\varepsilon > 0$ . Put

$$c = \max\{\alpha, \beta\}, \quad a = \frac{\min\{\alpha, \beta\}}{c} \in (0, 1).$$

Every component of  $\sigma_\varepsilon(T_n)$  is contractible, and

$$\sigma_\varepsilon(T_n) \text{ is connected} \iff \varepsilon > c\gamma_n(a).$$

The barriers  $c\gamma_n(a)$  decrease strictly with  $n$ , and the first connected dimension equals  $N_c(a, \varepsilon/c)$ . In particular, all bounds and asymptotic conclusions of theorem 2.1 transfer after the replacement  $\varepsilon \mapsto \varepsilon/c$ .

*Proof.* If  $\alpha < \beta$ , then  $T_n = dI_n + \beta A_n(\alpha/\beta)$ . If  $\alpha > \beta$ , then

$$T_n = J_n [dI_n + \alpha A_n(\beta/\alpha)] J_n,$$

where  $J_n$  is the reversal permutation matrix. Unitary conjugation preserves pseudospectra, while translation and positive scaling give

$$\sigma_\varepsilon(dI_n + cA) = d + c\sigma_{\varepsilon/c}(A).$$

The assertion follows from theorem 2.1, because affine maps of the plane preserve connectedness and contractibility.  $\square$

### 3. Vertical monotonicity and topological reduction

The topological reduction rests on a monotonicity property that makes every pseudospectral component connection visible on the real axis.

**Lemma 3.1** (Vertical monotonicity). *Let  $n \geq 2$  and  $0 < a < 1$ . For every real  $x$ ,*

$$y \longmapsto s_{\min}((x + iy)I_n - A_n), \quad y \geq 0,$$

*is nondecreasing.*

*Proof.* Put  $Y = y^2$ , and, for  $t \geq 0$ , define

$$\begin{aligned} \mathcal{P}_n(Y, t) &= ((x + i\sqrt{Y})I_n - A_n)^*((x + i\sqrt{Y})I_n - A_n) - t^2 I_n, \\ F_n(Y, t) &= \det \mathcal{P}_n(Y, t). \end{aligned}$$

Theorem A.1 proves the finite-dimensional implication

$$\mathcal{P}_n(Y, t) > 0 \implies \partial_Y F_n(Y, t) > 0. \quad (3.1)$$

Finally fix  $Y_0 \geq 0$ . When the least singular value at  $Y_0$  equals zero, monotonicity follows from nonnegativity of singular values. Otherwise take  $0 \leq t < s_{\min}((x + i\sqrt{Y_0})I_n - A_n)$ . On the maximal interval to the right of  $Y_0$  on which  $\mathcal{P}_n(Y, t) > 0$ , implication (3.1) makes its determinant strictly increasing. At any finite right endpoint, continuity of Hermitian eigenvalues would produce a positive-semidefinite singular matrix with determinant zero. Strict increase from the positive determinant at  $Y_0$  precludes such an endpoint. Thus  $\mathcal{P}_n(Y, t) > 0$  for every  $Y \geq Y_0$ . Letting  $t$  increase to the least singular value at  $Y_0$  proves the lemma.  $\square$

Consequently, at fixed real part, every vertical section of the strict pseudospectrum is an interval symmetric about the real axis. We now turn this analytic fact into a complete topological reduction.

#### 3.1. Topological reduction

The other ingredient is the standard component property for arbitrary matrices [6].

**Lemma 3.2.** *Let  $n \geq 2$ ,  $0 < a < 1$ , and  $\varepsilon > 0$ . Every connected component of  $\sigma_\varepsilon(A_n)$  contains an eigenvalue of  $A_n$ .*

*Proof.* The estimate

$$s_{\min}(zI_n - A_n) \geq |z| - \|A_n\|_2$$

shows that the pseudospectrum is bounded. Let  $\mathcal{C}$  be one of its components. Any eigenvalue  $\lambda \in \bar{\mathcal{C}}$  has, by continuity of the least singular value, a ball contained in the pseudospectrum. That ball meets  $\mathcal{C}$ , so maximality of connected components places  $\lambda$  in  $\mathcal{C}$ .

To prove that  $\mathcal{C}$  contains an eigenvalue, assume instead that it lies entirely in the resolvent set. The preceding observation then places  $\bar{\mathcal{C}}$  in the resolvent set as well. On a neighborhood of  $\bar{\mathcal{C}}$ , the function  $\mathbf{u}(z) = \|(zI_n - A_n)^{-1}\|_2$  is continuous and subharmonic [13]. Moreover,  $\mathbf{u} \leq \varepsilon^{-1}$  on  $\partial\mathcal{C}$ : a boundary point with  $\mathbf{u} > \varepsilon^{-1}$  would have a neighborhood in the pseudospectrum that meets  $\mathcal{C}$ , and hence would belong to  $\mathcal{C}$ . The maximum principle gives  $\mathbf{u} \leq \varepsilon^{-1}$  throughout  $\mathcal{C}$ , whereas the definition of the pseudospectrum gives  $\mathbf{u} > \varepsilon^{-1}$  there, a contradiction.  $\square$

Recall from (1.3) that all eigenvalues of  $A_n$  are real and that their convex hull is  $I_n^{\text{sp}}$ . Put  $\Omega = \sigma_\varepsilon(A_n)$ . Because  $A_n$  is real, the least singular value is invariant under complex conjugation. Theorem 3.1 therefore implies that

$$\mathcal{H}_t(x + iy) = x + i(1 - t)y, \quad 0 \leq t \leq 1,$$

maps  $\Omega$  into itself. If  $\mathcal{C}$  is a component of  $\Omega$ , the path  $t \mapsto \mathcal{H}_t(z)$  starts in  $\mathcal{C}$  and hence remains there. Thus  $\mathcal{H}_t|_{\mathcal{C}}$  is a strong deformation retraction onto  $\mathcal{C} \cap \mathbb{R}$ . This intersection is nonempty by Theorem 3.2 and connected as the image  $\mathcal{H}_1(\mathcal{C})$ , so it is an interval. Therefore every component is contractible.

If  $\Omega$  is connected, its real intersection is an interval containing every eigenvalue, and therefore contains their convex hull  $I_n^{\text{sp}}$ . Conversely, if  $I_n^{\text{sp}} \subset \Omega$ , it lies in a component containing every eigenvalue. Theorem 3.2 assigns an eigenvalue to every component, so this component is all of  $\Omega$ . Compactness of  $I_n^{\text{sp}}$  now gives

$$\Omega \text{ is connected} \iff I_n^{\text{sp}} \subset \Omega \iff \max_{x \in I_n^{\text{sp}}} g_n(x) < \varepsilon \iff \gamma_n < \varepsilon. \quad (3.2)$$

The strict inequality reflects the open definition of the pseudospectrum. At  $\varepsilon = \gamma_n$ , a maximizing barrier point remains excluded.

#### 4. Strict finite-size comparison of real-gap barriers

The final barrier can occur in any inter-eigenvalue gap, and the identity of the central gap changes with parity. A global finite-size comparison must therefore control every gap. Persymmetry turns singular values into signed eigenvalue branches, folded Chebyshev coordinates parametrize those

branches, scale inequalities compare noncentral gaps, and a separate interlacing argument handles the parity-changing central gaps. We first define the local real-gap barriers. Write the eigenvalues in decreasing order as

$$\lambda_{n,k} = 2r \cos \frac{k\pi}{n+1}, \quad 1 \leq k \leq n,$$

and call

$$G_{n,k} = [\lambda_{n,k+1}, \lambda_{n,k}], \quad \gamma_{n,k} = \max_{x \in G_{n,k}} g_n(x), \quad 1 \leq k < n,$$

the closed spectral gaps and their heights. Continuity guarantees that the maximum on each compact gap is attained, while the spectral endpoints have height zero. Every interior gap point belongs to the resolvent set, so  $xI_n - A_n$  is invertible there and  $g_n(x) > 0$ . Consequently every gap height is positive and every maximizer lies in the open gap. Hence

$$\gamma_n = \max_{1 \leq k < n} \gamma_{n,k}. \quad (4.1)$$

**Lemma 4.1** (Strict barrier comparison). *For every  $n \geq 2$ ,*

$$\gamma_{n+1} < \gamma_n.$$

Let

$$J_n = [\mathbf{1}_{\{i+j=n+1\}}]_{i,j=1}^n$$

be the reversal matrix. Persymmetry supplies the signed spectral representation needed for the finite-size comparison:  $A_n J_n = J_n A_n^T$ , so  $B_n(x) = (xI_n - A_n)J_n$  is real symmetric. Moreover,

$$(xI_n - A_n - sJ_n)J_n = B_n(x) - sI_n.$$

Since  $J_n$  is orthogonal, the absolute eigenvalues of  $B_n(x)$  are exactly the singular values of  $xI_n - A_n$ , including multiplicity. Consequently,

$$s_{\min}(xI_n - A_n) = \min\{|s| : \det(xI_n - A_n - sJ_n) = 0\}. \quad (4.2)$$

Let  $q_n(x, s) = \det(xI_n - A_n - sJ_n)$ .

**Lemma 4.2** (The middle signed branch). *Write the eigenvalues of  $B_n(x)$  in decreasing order as  $\alpha_1(x) \geq \dots \geq \alpha_n(x)$ , and abbreviate  $\lambda_k = \lambda_{n,k}$ . If  $\lambda_{j+1} < x < \lambda_j$ , then*

$$s_{\min}(xI_n - A_n) = |s_*(x)|, \quad \text{sgn}(s_*(x)) = (-1)^j,$$

where

$$s_*(x) = \begin{cases} \alpha_m(x), & n = 2m, \\ \alpha_{m+1}(x), & n = 2m + 1. \end{cases}$$

*Proof.* Let  $A_n^{\text{sym}} = r(V_n + V_n^T)$ . The identities

$$\begin{aligned} A_n &= \mathcal{G}_n(r) A_n^{\text{sym}} \mathcal{G}_n(r)^{-1}, \\ \mathcal{G}_n(r)^{-1} J_n &= r^{-(n-1)} J_n \mathcal{G}_n(r) \end{aligned}$$

give

$$B_n(x) = r^{-(n-1)} \mathcal{G}_n(r) (xI_n - A_n^{\text{sym}}) J_n \mathcal{G}_n(r). \quad (4.3)$$

The sine eigenvector of  $A_n^{\text{sym}}$  belonging to  $\lambda_k$  is an eigenvector of  $J_n$  with eigenvalue  $(-1)^{k+1}$ . Hence (4.3) and Sylvester's law of inertia show that  $B_n(x)$  has the same inertia as the diagonal matrix with entries

$$(-1)^{k+1}(x - \lambda_k), \quad 1 \leq k \leq n.$$

Counting signs gives

$$\nu_+(B_{2m}(x)) = \begin{cases} m, & j \text{ even,} \\ m-1, & j \text{ odd,} \end{cases} \quad \nu_+(B_{2m+1}(x)) = \begin{cases} m+1, & j \text{ even,} \\ m, & j \text{ odd.} \end{cases} \quad (4.4)$$

First let  $n = 2m$ . Permute the coordinates into the two groups

$$(1, \dots, m), \quad (2m, 2m-1, \dots, m+1).$$

Here  $\mathbf{e}_j$  denotes the  $j$ th Euclidean coordinate vector in the ambient space. In these coordinates

$$\begin{aligned} B_{2m}(x) &= \hat{P}_m(x) - \mathcal{R}_{\text{bd}}, \\ \hat{P}_m(x) &= \begin{bmatrix} 0 & P_m(x) \\ P_m(x)^T & 0 \end{bmatrix}, \\ \mathcal{R}_{\text{bd}} &= \begin{bmatrix} \mathbf{e}_m \mathbf{e}_m^T & 0 \\ 0 & a \mathbf{e}_m \mathbf{e}_m^T \end{bmatrix}, \end{aligned} \quad (4.5)$$

where  $P_m(x) = xI_m - A_m$ . Thus  $\mathcal{R}_{\text{bd}} \geq 0$  and  $\text{rank } \mathcal{R}_{\text{bd}} = 2$ . For  $\Gamma = \text{diag}(I_m, -I_m)$ ,

$$\tilde{B} = -\Gamma B_{2m}(x) \Gamma = \hat{P}_m(x) + \mathcal{R}_{\text{bd}} = B_{2m}(x) + 2\mathcal{R}_{\text{bd}}.$$

If  $\tilde{\alpha}_1 \geq \dots \geq \tilde{\alpha}_{2m}$  are the eigenvalues of  $\tilde{B}$ , then

$$\tilde{\alpha}_k = -\alpha_{2m+1-k}. \quad (4.6)$$

Positive-semidefinite rank-two interlacing gives

$$\tilde{\alpha}_k \geq \alpha_k \geq \tilde{\alpha}_{k+2}, \quad 1 \leq k \leq 2m-2. \quad (4.7)$$

If  $j$  is even, the only candidates for minimum modulus have moduli  $\alpha_m$  and  $-\alpha_{m+1} = \tilde{\alpha}_m$ , and  $\alpha_m \leq \tilde{\alpha}_m$ . If  $j$  is odd and  $m \geq 2$ , the two candidates have moduli  $\alpha_{m-1}$  and  $-\alpha_m = \tilde{\alpha}_{m+1}$ ; taking  $k = m-1$  in (4.7) gives  $\alpha_{m-1} \geq \tilde{\alpha}_{m+1}$ . For  $m = 1$ , both eigenvalues in the sole gap are negative, so the larger one  $\alpha_1$  has minimum modulus. Thus  $\alpha_m$  always attains the minimum modulus, and its sign follows from (4.4). Equality in an interlacing comparison is harmless: it only gives a multiple least singular value.

Now let  $n = 2m + 1$ . Group the first and last  $m$  coordinates as above and leave the middle coordinate last. Then

$$B_{2m+1}(x) = \begin{bmatrix} \hat{P}_m(x) & \mathbf{w} \\ \mathbf{w}^T & x \end{bmatrix}, \quad \mathbf{w} = \begin{bmatrix} -\mathbf{e}_m \\ -a\mathbf{e}_m \end{bmatrix}. \quad (4.8)$$

In an open spectral gap of  $A_{2m+1}$ ,  $P_m(x)$  is nonsingular. Indeed, the explicit path spectrum gives

$$\text{spec}(A_m) = \left\{ 2r \cos \frac{\ell\pi}{m+1} : 1 \leq \ell \leq m \right\} = \{ \lambda_{2m+1, 2\ell} : 1 \leq \ell \leq m \},$$

so its zeros are exactly the even-indexed eigenvalues of  $A_{2m+1}$ , none of which lies in an open spectral gap. The eigenvalues of the principal submatrix  $\hat{P}_m(x)$  are consequently

$$s_1(P_m) \geq \cdots \geq s_m(P_m) > 0 > -s_m(P_m) \geq \cdots \geq -s_1(P_m).$$

Cauchy interlacing gives

$$\alpha_m \geq s_m(P_m) \geq \alpha_{m+1} \geq -s_m(P_m) \geq \alpha_{m+2}.$$

If  $j$  is even, (4.4) and interlacing give

$$0 < \alpha_{m+1} \leq s_m(P_m) \leq -\alpha_{m+2};$$

if  $j$  is odd, they give

$$0 < -\alpha_{m+1} \leq s_m(P_m) \leq \alpha_m.$$

Every other eigenvalue is farther from zero. Hence  $\alpha_{m+1}$  attains the minimum modulus, with sign prescribed by (4.4); equality again only means multiplicity. At a spectral endpoint the zero is simple because  $A_n$  is similar to the Jacobi matrix  $A_n^{\text{sym}}$ , whose eigenvalues are simple. Thus the selected ordered branch extends continuously to zero at both ends of the closed gap. This proves the assertion.  $\square$

We use folded coordinates for the noncentral gaps and a separate interlacing argument for the parity-changing central gaps.

*Folded continuants*

Let  $U_j$  be the Chebyshev polynomial of the second kind, with  $U_{-1} = 0$ . Put

$$C_0 = x^2 + (1 - a)^2 - s^2, \quad \zeta_s = \frac{1 + a^2 + (1 + a)s}{2a},$$

and define the folded discriminant and recurrence coefficient by

$$\begin{aligned} \Delta_{\text{fold}} &= ((1 + a - x)^2 - s^2)((1 + a + x)^2 - s^2), \\ \mathfrak{b} &= (1 + a^2)x^2 + 2as^2 + 2a^2 - 2a - 2a^3. \end{aligned}$$

The global recurrence polynomial is

$$\begin{aligned} \Pi(z) &:= 1 - C_0z + \mathfrak{b}z^2 - a^2C_0z^3 + a^4z^4, \\ u, v \in \mathbb{R}, \quad 2a(u + v) &= C_0, \quad a^2(4uv + 2) = \mathfrak{b}, \\ \implies \Pi(z) &= (1 - 2auz + a^2z^2)(1 - 2avz + a^2z^2). \end{aligned} \tag{4.9}$$

On a real sheet with  $\Delta_{\text{fold}} \geq 0$ , we additionally put

$$\xi_{\pm} = \frac{C_0 \pm \sqrt{\Delta_{\text{fold}}}}{4a}.$$

Then the pair  $(u, v) = (\xi_+, \xi_-)$  satisfies the two symmetric relations in (4.9):

$$\xi_+ + \xi_- = \frac{C_0}{2a}, \quad \mathfrak{b} = a^2(4\xi_+\xi_- + 2).$$

We use the root presentation on these real sheets and the polynomial  $\Pi$  itself for the global transfer and generating identities. The complete five-minor transfer, characteristic-polynomial calculation, boundary residuals, and generating functions are given in appendix Appendix B. Their parity-unified conclusion is the part needed in the main argument. For a differentiable real function  $f$ , write

$$[f]_{u,v} := \begin{cases} \frac{f(u) - f(v)}{u - v}, & u \neq v, \\ f'(u), & u = v \end{cases}.$$

Whenever the real pair  $u, v$  satisfies the two symmetric relations in (4.9), the divided-difference calculation in appendix Appendix C gives

$$q_{2m}(x, s) = a^m [\mathfrak{f}_m]_{u,v}, \quad \mathfrak{f}_m(t) = (t - \zeta_s)(U_m(t) + U_{m-1}(t)), \tag{4.10}$$

and

$$q_{2m+1}(x, s) = a^m [\mathfrak{g}_m]_{u,v}, \quad \mathfrak{g}_m(t) = \left( (x-s)t - \frac{x(1+a^2) + 2as}{2a} \right) U_m(t). \quad (4.11)$$

The recurrence and generating identities are polynomial and therefore global. The real-pair formulas are used only after the two symmetric relations have been checked; at  $u = v$ , the displayed derivative is the collision convention. For  $\Delta_{\text{fold}} < 0$ , the argument stays at the polynomial level and avoids any complex square-root branch.

These formulas replace the singular-value problem by a signed scalar branch with parity already resolved. The next step identifies a geometric parametrization of that branch.

*Half-angle chord parametrization*

Put  $h = -\frac{1}{2} \log a > 0$ , so that  $a = e^{-2h}$ , and set

$$\mathfrak{s}_h = \sinh h, \quad \mathfrak{c}_h = \cosh h.$$

For  $K \geq 2$  and  $0 \leq \theta \leq \pi$ , define

$$\Phi_K(\theta) = \sqrt{\mathfrak{s}_h^2 + \sin^2 \theta} \frac{|\sin(K\theta)|}{\sin \theta}, \quad (4.12)$$

where the quotient is interpreted continuously at  $\theta = 0, \pi$ , and

$$\Psi_K(\eta) = \sqrt{\mathfrak{s}_h^2 - \sinh^2 \eta} \frac{\sinh(K\eta)}{\sinh \eta}, \quad 0 \leq \eta \leq h,$$

with the limiting value at  $\eta = 0$ . The following half-angle reduction also fixes the sign convention. Put

$$y = \cos \theta, \quad z = \cosh \eta, \quad \mathfrak{p}_\theta = \sqrt{\mathfrak{c}_h^2 - y^2}, \quad \mathfrak{p}_\eta = \sqrt{\mathfrak{c}_h^2 - z^2},$$

and let  $L_0 = 2r/\mathfrak{c}_h$ . If  $x = L_0 y z$ ,  $s = L_0 \mathfrak{p}_\theta \mathfrak{p}_\eta > 0$ , and  $\tau = \varsigma s$  with  $\varsigma \in \{-1, 1\}$ , then direct expansion gives

$$\begin{aligned} (1+a+x)^2 - s^2 &= 4a(z+y)^2, & (1+a-x)^2 - s^2 &= 4a(z-y)^2, \\ u = 2z^2 - 1, & v = 2y^2 - 1, & \zeta_\tau &= 2\mathfrak{c}_h^2 - 1 + 2\varsigma \mathfrak{p}_\theta \mathfrak{p}_\eta. \end{aligned} \quad (4.13)$$

The real pair  $(u, v)$  satisfies the two symmetric relations in (4.9). For  $K = 2m + 1$ , use

$$U_m(2t^2 - 1) + U_{m-1}(2t^2 - 1) = U_{K-1}(t);$$

for  $K = 2m + 2$ , use

$$U_m(2t^2 - 1) = \frac{U_{K-1}(t)}{2t}.$$

Substitution of this pair in (4.10) and (4.11), respectively, reduces either equation  $q_{K-1}(x, \tau) = 0$  to

$$\mathbf{p}_\theta U_{K-1}(y) = \varsigma \mathbf{p}_\eta U_{K-1}(z). \quad (4.14)$$

In the even-fold case the intermediate factorization is

$$(\mathbf{p}_\eta + \varsigma \mathbf{p}_\theta) (\mathbf{p}_\eta U_{K-1}(z) - \varsigma \mathbf{p}_\theta U_{K-1}(y)) = 0.$$

The first factor could vanish only if  $\varsigma = -1$  and  $\mathbf{p}_\eta = \mathbf{p}_\theta$ , which would imply  $y^2 = z^2$ ; the distinct-variable condition therefore leaves precisely (4.14). In the odd-fold case the reduction also contains the factor  $\mathbf{p}_\eta y + \varsigma \mathbf{p}_\theta z$ . On a distinct-variable sheet this factor is nonzero: if it vanished, squaring would give

$$(\mathbf{c}_h^2 - z^2)y^2 = (\mathbf{c}_h^2 - y^2)z^2,$$

and hence  $y^2 = z^2$ . Thus the factor is nonzero whenever the folded variables are distinct. On the outer lobe  $U_{K-1}(z) > 0$ , whereas, for  $j\pi/K < \theta < (j+1)\pi/K$ ,

$$\text{sgn}(U_{K-1}(\cos \theta)) = \text{sgn}(\sin(K\theta)) = (-1)^j.$$

Moreover,  $\mathbf{p}_\theta |U_{K-1}(y)| = \Phi_K(\theta)$  and  $\mathbf{p}_\eta U_{K-1}(z) = \Psi_K(\eta)$ . Thus (4.14) is equivalent to  $\varsigma = (-1)^j$  and

$$\Phi_K(\theta) = \Psi_K(\eta), \quad (4.15)$$

with

$$s = \mathcal{S}_{\text{hyp}}(\theta, \eta) := \frac{2r}{\mathbf{c}_h} \sqrt{(\mathbf{s}_h^2 + \sin^2 \theta)(\mathbf{s}_h^2 - \sinh^2 \eta)}, \quad x = \frac{2r}{\mathbf{c}_h} \cos \theta \cosh \eta. \quad (4.16)$$

Consequently  $q_{K-1}(x, (-1)^j s) = 0$ . Conversely, for distinct real folded variables, their sum and product determine

$$x^2 = L_0^2 y^2 z^2, \quad s^2 = L_0^2 (\mathbf{c}_h^2 - y^2)(\mathbf{c}_h^2 - z^2).$$

Thus, for distinct real variables and a fixed sign of  $x$ , the preceding steps are reversible, and the two chord relations are equivalent to the corresponding signed equation. The determinant identities themselves extend to a folded-variable collision by the continuous divided-difference convention, but we

invoke the chord equivalence only for distinct variables; the selected middle branch is proved below to be collision-free. Here  $K-1$  is the matrix size. If the exterior argument has crossed through 1, write it as  $z = \cos \phi$ ,  $0 \leq \phi < \pi/K$ . Then the chord equation is  $\Phi_K(\theta) = \Phi_K(\phi)$ , and the second factor in the formula for  $s$  is  $\mathfrak{s}_h^2 + \sin^2 \phi$ . We denote the resulting value by

$$\mathcal{S}_{\text{ell}}(\theta, \phi) = \frac{2r}{\mathfrak{c}_h} \sqrt{(\mathfrak{s}_h^2 + \sin^2 \theta)(\mathfrak{s}_h^2 + \sin^2 \phi)}.$$

#### *Branch classification and outer-lobe geometry*

It remains to identify which chords realize the middle branch of Theorem 4.2. Set

$$\mathcal{W}_K(u) = \sqrt{\mathfrak{c}_h^2 - u^2} |U_{K-1}(u)|, \quad |u| \leq \mathfrak{c}_h. \quad (4.17)$$

At  $u = \pm \mathfrak{c}_h$ , this denotes the continuous endpoint value 0. Then

$$\Phi_K(\theta) = \mathcal{W}_K(\cos \theta), \quad \Psi_K(\eta) = \mathcal{W}_K(\cosh \eta).$$

If  $u_{\ell,K}^{(0)} = \cos \frac{\ell\pi}{K}$ , then, away from the zeros,

$$\frac{d}{du} \log \mathcal{W}_K(u) = \sum_{\ell=1}^{K-1} \frac{1}{u - u_{\ell,K}^{(0)}} - \frac{u}{\mathfrak{c}_h^2 - u^2},$$

and its derivative is

$$-\sum_{\ell=1}^{K-1} \frac{1}{(u - u_{\ell,K}^{(0)})^2} - \frac{\mathfrak{c}_h^2 + u^2}{(\mathfrak{c}_h^2 - u^2)^2} < 0. \quad (4.18)$$

At each finite nodal endpoint the logarithmic derivative tends to the appropriate signed infinity, and it tends to  $-\infty$  as  $u \uparrow \mathfrak{c}_h$ . Thus there is exactly one lobe maximum in every nodal interval, including the outer interval  $\mathcal{J}_K^{\text{out}} = (\cos \frac{\pi}{K}, \mathfrak{c}_h)$ . Let  $u_K^{\text{max}}$  be the unique maximizer on  $\mathcal{J}_K^{\text{out}}$ . For every  $0 \leq \ell < \mathcal{W}_K(u_K^{\text{max}})$ , there is a unique solution

$$z_K(\ell) \in (u_K^{\text{max}}, \mathfrak{c}_h], \quad \mathcal{W}_K(z_K(\ell)) = \ell;$$

we call it the *endpoint-side outer solution*. If  $z_K(\ell) \geq 1$ , write it as  $\cosh \eta$ ,  $0 \leq \eta \leq h$ ; if it is at most 1, write it as  $\cos \phi$ ,  $0 \leq \phi < \pi/K$ . These are two coordinates on the same outer lobe, joined at  $z = 1$ .

For a positive-half gap and  $\theta \in (j\pi/K, (j+1)\pi/K)$ ,  $\theta \leq \pi/2$ , a  $j$ -chord is the pair

$$y = \cos \theta, \quad z = z_K(\mathcal{W}_K(y)), \quad \mathcal{W}_K(y) = \mathcal{W}_K(z),$$

together with the coordinate-free values

$$x = L_0 y z, \quad s = L_0 \sqrt{(\mathfrak{c}_h^2 - y^2)(\mathfrak{c}_h^2 - z^2)};$$

its signed root is  $(-1)^j s$ . The term endpoint-side always refers to the part of the outer lobe adjacent to  $\mathfrak{c}_h$ , independently of whether it is described by the hyperbolic or elliptic coordinate.

Reflection reduces the branch classification to a positive-half gap, so  $j \geq 1$  and  $\theta \leq \pi/2$ . Put  $\phi_{\text{rem}} = \theta - j\pi/K \in (0, \pi/K)$ . Since  $|\sin(K\phi_{\text{rem}})| = |\sin(K\theta)|$  and

$$\mathcal{A}_h(t) = \frac{\sqrt{\mathfrak{s}_h^2 + \sin^2 t}}{\sin t}$$

is strictly decreasing on  $(0, \pi/2]$ ,

$$\Phi_K(\theta) < \Phi_K(\phi_{\text{rem}}) \leq \max_{u \in \mathcal{J}_K^{\text{out}}} \mathcal{W}_K(u). \quad (4.19)$$

Thus the endpoint-side outer solution exists for every inner parameter.

At  $x = \lambda_{K-1,k} = 2r \cos \frac{k\pi}{K}$ ,  $s = 0$ , the two folded variables are  $\cos \frac{2k\pi}{K}$  and  $\cosh(2h)$ , or, in half variables,  $\cos \frac{k\pi}{K}$  and  $\mathfrak{c}_h$ . Moreover, zero is a simple eigenvalue of  $B_{K-1}(\lambda_{K-1,k})$ : the matrix  $\lambda_{K-1,k} I_{K-1} - A_{K-1}$  has rank  $K-2$ , and  $J_{K-1}$  is invertible. The implicit-function theorem therefore gives, near either gap endpoint, a real continuation of the middle signed root and real half variables  $y = \cos \theta$  and  $z$ .

#### *Chord-sheet exhaustion*

By theorem D.1, the ordered middle branch in every positive gap is exactly the selected endpoint-side chord sheet. In particular, its signed root is  $(-1)^j s$ , its magnitude is the least singular value, and its folded variables are distinct in the open gap. The same result covers the positive half of the central gap when  $K$  is odd.

**Lemma 4.3** (Noncentral gap comparison). *Let  $N \geq 3$  and  $j \geq 1$  be integers, put  $M = N + 1$ , and suppose*

$$\frac{(j+1)\pi}{N} \leq \frac{\pi}{2}. \quad (4.20)$$

*For every  $x \in G_{M-1,j}^\circ$ , there is  $\hat{x} \in G_{N-1,j}^\circ$  such that*

$$g_{N-1}(\hat{x}) > g_{M-1}(x).$$

*Proof.* Fix  $x \in G_{M-1,j}^\circ$ . By theorem D.1, its middle branch has a hyperbolic or elliptic chord representation with

$$\theta \in \left( \frac{j\pi}{M}, \frac{(j+1)\pi}{M} \right).$$

Set  $\kappa = N/M$  and  $\theta_0 = \theta/\kappa$ . The hypotheses place  $\theta_0$  in the corresponding  $N$ -nodal interval, with every trigonometric argument below in  $(0, \pi/2]$ .

First suppose  $\Phi_M(\theta) = \Psi_M(\eta)$ . Concavity of sine gives

$$\frac{\Phi_N(\theta_0)}{\Phi_M(\theta)} = \frac{\mathcal{A}_h(\theta_0)}{\mathcal{A}_h(\theta)} > \kappa. \quad (4.21)$$

Indeed,  $\theta = \kappa\theta_0$  and strict concavity gives  $\sin \theta > \kappa \sin \theta_0$ ; hence

$$\mathcal{A}_h(\theta_0)^2 - \kappa^2 \mathcal{A}_h(\theta)^2 = (1 - \kappa^2) + \mathfrak{s}_h^2 \left( \frac{1}{\sin^2 \theta_0} - \frac{\kappa^2}{\sin^2 \theta} \right) > 0.$$

On the other hand,

$$\frac{\Psi_N(\eta)}{\Psi_M(\eta)} = \frac{\sinh(N\eta)}{\sinh(M\eta)} \leq \kappa, \quad (4.22)$$

with strict inequality for  $\eta > 0$ , by convexity of  $\sinh$  on the positive half-axis. Hence  $\Phi_N(\theta_0) > \Psi_N(\eta)$ . Since  $\Phi_N$  vanishes at the right endpoint  $(j+1)\pi/N$ , its rightmost solution  $\hat{\theta}$  of

$$\Phi_N(\hat{\theta}) = \Psi_N(\eta)$$

exists by the intermediate value theorem and satisfies  $\hat{\theta} > \theta_0 > \theta$ . Strict log-concavity of  $\mathcal{W}_N$  makes it the unique solution on the descending half of that lobe. The same exterior variable is endpoint-side for the  $N$ -problem as well. Indeed, for  $u = \cosh \eta > 1$ ,

$$\frac{d}{du} \log \mathcal{W}_M(u) - \frac{d}{du} \log \mathcal{W}_N(u) = \frac{M \coth(M\eta) - N \coth(N\eta)}{\sinh \eta} > 0.$$

The numerator is positive because, for  $\eta > 0$ ,

$$\frac{d}{dt} (t \coth(t\eta)) = \frac{\sinh(t\eta) \cosh(t\eta) - t\eta}{\sinh^2(t\eta)} > 0.$$

The endpoint-side branch is characterized by a nonpositive logarithmic derivative. The displayed difference therefore makes the  $N$ -derivative strictly negative. The case  $\eta = 0$  follows by continuity. The chord classification above says that the constructed value  $\mathcal{S}_{\text{hyp}}(\hat{\theta}, \eta)$  is the least singular value at its

constructed abscissa. Formula (4.16) and the strict increase of sine on the interval now give

$$\mathcal{S}_{\text{hyp}}(\widehat{\theta}, \eta) > \mathcal{S}_{\text{hyp}}(\theta, \eta). \quad (4.23)$$

Suppose next that the exterior endpoint is elliptic, so that  $\Phi_M(\theta) = \Phi_M(\phi)$ ,  $0 < \phi < \pi/M$ . (The case  $\phi = 0$  is the limiting hyperbolic case already covered.) Put

$$\theta_0 = \theta/\kappa, \quad \phi_0 = \phi/\kappa.$$

The ratio

$$\mathcal{A}_{N,M}^\Phi(t) = \frac{\Phi_N(t/\kappa)}{\Phi_M(t)} = \frac{\mathcal{A}_h(t/\kappa)}{\mathcal{A}_h(t)}$$

is strictly increasing while  $0 < t < t/\kappa \leq \pi/2$ . Indeed,

$$\frac{d}{dt} \log \mathcal{A}_{N,M}^\Phi(t) = \mathfrak{d}_h(t) - \kappa^{-1} \mathfrak{d}_h(t/\kappa) > 0, \quad \mathfrak{d}_h(t) = \frac{\mathfrak{s}_h^2 \cot t}{\mathfrak{s}_h^2 + \sin^2 t},$$

because

$$t \mathfrak{d}_h(t) = (t \cot t) \frac{\mathfrak{s}_h^2}{\mathfrak{s}_h^2 + \sin^2 t}$$

is strictly decreasing: both positive factors decrease on  $(0, \pi/2)$ , and the second decreases strictly. Since  $\theta > \pi/M > \phi$ , it follows that  $\Phi_N(\theta_0) > \Phi_N(\phi_0)$ . The rightmost solution of

$$\Phi_N(\widehat{\theta}) = \Phi_N(\phi_0)$$

therefore has  $\widehat{\theta} > \theta_0$ . Its exterior variable is endpoint-side. Indeed, if  $\Lambda_K(t) = (\log \Phi_K)'(t)$ , then on the endpoint-side part of the first outer lobe  $\Lambda_M(\phi) \geq 0$ , and differentiation of  $\Phi_N(t/\kappa) = \mathcal{A}_{N,M}^\Phi(t) \Phi_M(t)$  gives

$$\kappa^{-1} \Lambda_N(t/\kappa) = \Lambda_M(t) + \frac{(\mathcal{A}_{N,M}^\Phi)'(t)}{\mathcal{A}_{N,M}^\Phi(t)} > 0.$$

Because  $\Lambda_K(\phi) = -\sin \phi (\log \mathcal{W}_K)'(\cos \phi)$ , positivity of  $\Lambda_N(\phi_0)$  is exactly the endpoint-side condition for the  $N$ -outer solution. The one-maximum property now fixes the label, and the chord classification says again that  $\mathcal{S}_{\text{ell}}(\widehat{\theta}, \phi_0)$  is the least singular value at the constructed abscissa. Here  $\widehat{\theta} > \theta_0 > \theta$  and  $\phi_0 > \phi$ , with all four angles below  $\pi/2$ . Thus both sine factors in the elliptic version of (4.16) have increased, at least one strictly, and hence

$$\mathcal{S}_{\text{ell}}(\widehat{\theta}, \phi_0) > \mathcal{S}_{\text{ell}}(\theta, \phi). \quad (4.24)$$

Equations (4.23) and (4.24), together with the coordinate formula for  $x$ , place the constructed abscissa  $\widehat{x}$  in  $G_{N-1,j}^\circ$ . Chord exhaustion identifies the original and constructed heights with  $g_{M-1}(x)$  and  $g_{N-1}(\widehat{x})$ , respectively, proving the claim.  $\square$

For  $m \geq 1$ , let

$$c_m = s_{\min}(aI_m + V_m). \quad (4.25)$$

By reflection, the two gaps adjacent to the zero eigenvalue of  $A_{2m+1}$  have the same height; denote it by

$$d_m = \max_{0 \leq x \leq 2r \sin \frac{\pi}{2m+2}} s_{\min}(xI_{2m+1} - A_{2m+1}). \quad (4.26)$$

**Lemma 4.4** (Central gap comparison). *For every  $m \geq 1$ ,*

$$c_m > d_m > c_{m+1}. \quad (4.27)$$

*Proof.* Put

$$\zeta = \frac{a + a^{-1}}{2}.$$

For every  $j \geq 1$ , there is a unique  $\rho_j > \cos \frac{\pi}{j+1}$  such that

$$aU_j(\rho_j) = U_{j-1}(\rho_j). \quad (4.28)$$

Indeed, the Christoffel–Darboux identity gives, above the largest zero of  $U_j$ ,

$$\frac{d}{dt} \frac{U_{j-1}(t)}{U_j(t)} = -\frac{2 \sum_{k=0}^{j-1} U_k(t)^2}{U_j(t)^2} < 0.$$

The ratio decreases from  $+\infty$  to 0, proving existence and uniqueness. A three-term determinant recurrence gives

$$\det \left( (aI_j + V_j)^*(aI_j + V_j) - (1 + a^2 - 2at)I_j \right) = a^j \left( U_j(t) - a^{-1}U_{j-1}(t) \right). \quad (4.29)$$

Thus the largest root  $\rho_j$  gives the least eigenvalue. Together with

$$U_j(t)^2 + U_{j-1}(t)^2 - 2tU_j(t)U_{j-1}(t) = 1,$$

equations (4.28) and (4.29) give

$$c_j^2 = 1 + a^2 - 2a\rho_j = \frac{1}{U_j(\rho_j)^2}, \quad \zeta - \rho_j = \frac{1}{2U_{j-1}(\rho_j)U_j(\rho_j)}. \quad (4.30)$$

Next we identify  $c_m$  as the exact height of the central gap of  $A_{2m}$ . Let

$$P_{\text{oe}} = [\mathbf{e}_1, \mathbf{e}_3, \dots, \mathbf{e}_{2m-1}, \mathbf{e}_2, \mathbf{e}_4, \dots, \mathbf{e}_{2m}]$$

be the permutation matrix that places odd coordinates first. Then

$$P_{\text{oe}}^T A_{2m} P_{\text{oe}} = \begin{bmatrix} 0 & I_m + aV_m^T \\ aI_m + V_m & 0 \end{bmatrix}.$$

Moreover,

$$(I_m + aV_m^T)^*(I_m + aV_m^T) - J_m(aI_m + V_m)^*(aI_m + V_m)J_m = (1 - a^2)\mathbf{e}_m\mathbf{e}_m^T \geq 0.$$

The singular values of the block off-diagonal matrix are the union of the singular values of its two off-diagonal blocks. The Gram inequality shows that  $s_{\min}(I_m + aV_m^T) \geq s_{\min}(aI_m + V_m) = c_m$ , and hence  $s_{\min}(A_{2m}) = c_m$ . To identify the maximum throughout the central gap, put  $K = 2m + 1$  and use the half variables

$$y = \cos \theta, \quad z = \cosh \eta \quad \text{or} \quad z = \cos \phi.$$

On the positive half of the central middle branch, the preceding chord classification gives

$$\mathcal{W}_K(y) = \mathcal{W}_K(z), \quad x = \frac{2r}{\mathbf{c}_h}yz, \quad s^2 = \frac{4a}{\mathbf{c}_h^2}(\mathbf{c}_h^2 - y^2)(\mathbf{c}_h^2 - z^2), \quad (4.31)$$

where  $0 \leq y < \cos \frac{m\pi}{K}$  is in the central lobe and  $z$  is the endpoint-side outer solution. The function  $\mathcal{W}_K$  is even, and strict log-concavity on the central lobe gives

$$\mathcal{W}_K(y) < \mathcal{W}_K(0) = \mathbf{c}_h \quad (y > 0).$$

Let

$$z_0 = \sqrt{\frac{1 + \rho_m}{2}}.$$

The half-angle identity and (4.28) give

$$U_{2m}(z_0) = U_m(\rho_m) + U_{m-1}(\rho_m) = (1 + a)U_m(\rho_m).$$

Together with (4.30), this yields

$$\mathcal{W}_K(z_0) = \mathbf{c}_h, \quad 4a(\mathbf{c}_h^2 - z_0^2) = 2a(\zeta - \rho_m) = c_m^2. \quad (4.32)$$

At  $x = 0$ , theorem 4.2 and the chord-exhaustion argument identify the nearest root with  $c_m$ , so  $z_0$  is the endpoint-side outer solution of level  $\mathbf{c}_h$ . If  $x \neq 0$ , the level in (4.31) is strictly below  $\mathbf{c}_h$ , and hence its endpoint-side solution satisfies  $z > z_0$ . Equations (4.31)–(4.32) then give

$$s^2 < 4a(\mathbf{c}_h^2 - z_0^2) = c_m^2.$$

Reflection gives the same strict bound on the other half of the gap. Therefore the central-gap height equals  $c_m$  and has the unique maximizer  $x = 0$ .

Set

$$\omega_m^e = \cos \frac{\pi}{m+1}, \quad \mathcal{V}_m(t) = \sqrt{(\zeta - t)(1 + t)} U_m(t).$$

For the central specialization, take  $K = 2m + 2$  and set

$$t_{\text{in}} = 2y^2 - 1 = \cos(2\theta), \quad t_{\text{out}} = 2z^2 - 1.$$

On the positive central gap,  $-1 < t_{\text{in}} < -\omega_m^e$ , while the endpoint-side outer variable satisfies  $\omega_m^e < t_{\text{out}} < \zeta$ , where  $\zeta = 2\mathfrak{c}_h^2 - 1$ . The identity  $U_{2m+1}(t) = 2tU_m(2t^2 - 1)$  gives

$$\begin{aligned} \mathcal{W}_{2m+2}(y)^2 &= (\zeta - t_{\text{in}})(1 + t_{\text{in}})U_m(t_{\text{in}})^2, \\ \mathcal{W}_{2m+2}(z)^2 &= (\zeta - t_{\text{out}})(1 + t_{\text{out}})U_m(t_{\text{out}})^2. \end{aligned}$$

Because  $\zeta + 1 = 2\mathfrak{c}_h^2$ , the formulas  $x = (2r/\mathfrak{c}_h)yz$  and  $s^2 = (4a/\mathfrak{c}_h^2)(\mathfrak{c}_h^2 - y^2)(\mathfrak{c}_h^2 - z^2)$  become the following chord parametrization:

$$\begin{aligned} -1 < t_{\text{in}} < -\omega_m^e, \quad \omega_m^e < t_{\text{out}} < \zeta, \quad \mathcal{V}_m(t_{\text{out}}) = |\mathcal{V}_m(t_{\text{in}})|, \\ x^2 &= \frac{2a(1 + t_{\text{out}})(1 + t_{\text{in}})}{\zeta + 1}, \quad s^2 = \frac{2a(\zeta - t_{\text{out}})(\zeta - t_{\text{in}})}{\zeta + 1}, \end{aligned} \quad (4.33)$$

where  $t_{\text{out}}$  is the descending, or largest, solution. To see this from (4.15), use

$$\mathfrak{s}_h^2 + \sin^2 \theta = \frac{\zeta - t_{\text{in}}}{2}, \quad \mathfrak{s}_h^2 - \sinh^2 \eta = \frac{\zeta - t_{\text{out}}}{2}.$$

Continuation through  $t_{\text{out}} = 1$  uses the elliptic coordinate  $z = \cos \phi$ .

Let  $\rho = \rho_m$ . The function  $\log \mathcal{V}_m$  is strictly concave on  $(\omega_m^e, \zeta)$ . If  $(\log \mathcal{V}_m)'(\rho) \geq 0$ , then  $\rho$  lies at or before the unique lobe maximum. The descending root lies at or after that maximum, so  $t_{\text{out}} \geq \rho$ . Now suppose  $(\log \mathcal{V}_m)'(\rho) < 0$ . We claim that  $\rho > a$ . If  $\rho \leq a < 1$ , then  $\rho < 1$ , hence  $\rho^2 - 1 \neq 0$ , and the following quotient is well defined:

$$\frac{U'_m(\rho)}{U_m(\rho)} = \frac{m\rho - (m+1)a}{\rho^2 - 1}$$

would give

$$\begin{aligned} (\log \mathcal{V}_m)'(\rho) &= \frac{(2m+2)a + 1 - (2m+1)\rho}{2(1 - \rho^2)} - \frac{1}{2(\zeta - \rho)} \\ &\geq \frac{1}{2(1 - \rho)} - \frac{1}{2(\zeta - \rho)} \geq 0, \end{aligned}$$

which contradicts the assumed negative derivative. Hence  $\rho > a$ . Writing  $t_{\text{in}} = -\cos \phi_c$ ,  $0 < \phi_c < \pi/(m+1)$ , gives

$$\begin{aligned} |\mathcal{V}_m(t_{\text{in}})|^2 &= \frac{(\zeta + \cos \phi_c) \sin^2((m+1)\phi_c)}{1 + \cos \phi_c} \\ &< \frac{\zeta + \omega_m^e}{1 + \omega_m^e} < \frac{1+a}{2a} < \frac{1+\rho}{2a} = \mathcal{V}_m(\rho)^2. \end{aligned}$$

Now  $\rho$  is on the descending side and  $|\mathcal{V}_m(t_{\text{in}})| < \mathcal{V}_m(\rho)$ , so the unique descending root again satisfies  $t_{\text{out}} > \rho$ . In either case,  $t_{\text{out}} \geq \rho$ , and  $t_{\text{in}} > -1$  makes the following inequality strict:

$$s^2 < 2a(\zeta - \rho) = c_m^2.$$

This holds along the whole central sheet, so  $d_m < c_m$ . Thus the odd-order central height lies strictly below the preceding even-order central height. To complete the alternating comparison, it remains to place it strictly above the next even-order height.

*Lower comparison  $d_m > c_{m+1}$ : decisive scalar estimate*

Put  $L = m+1 \geq 2$ , and abbreviate

$$\begin{aligned} \rho = \rho_L, \quad u_- = U_{L-1}(\rho), \quad u_+ = U_L(\rho), \quad \omega_L = \cos \frac{\pi}{2L}, \quad \Delta_\omega = 1 - \omega_L, \\ \mathfrak{D}_L = 1 + 2(\rho + \omega_L)u_-u_+, \quad \xi = \rho - \frac{\Delta_\omega}{\mathfrak{D}_L}. \end{aligned} \quad (4.34)$$

Thus  $a = u_-/u_+$  and  $c_L = 1/u_+$ .

The required inequality is

$$(1 + \omega_L)(1 + \xi)^2 U_{L-1}(\xi)^2 < (\rho + \omega_L) \mathfrak{D}_L. \quad (4.35)$$

Appendix Appendix E proves (4.35) for every  $L \geq 2$  from the Chebyshev identity and strict log-concavity, treating the three cases  $\rho > 1$ ,  $\rho = 1$ , and  $\rho < 1$ .

*Certification of the least singular value*

Since  $a > 0$ ,  $\Delta_\omega > 0$ ,  $\rho + \omega_L > 0$ , and  $\zeta + \omega_L > 0$ , let  $x_* > 0$  be the positive square root determined by

$$x_*^2 = \frac{2a\Delta_\omega(\rho + \omega_L)}{\zeta + \omega_L}, \quad c = c_L. \quad (4.36)$$

The algebraic certification is given in appendix Appendix F. Substitution of the distinct folded pair  $u = \xi$ ,  $v = -\omega_L$  into (4.11), followed by (4.35), proves

$$q_{2L-1}(x_*, c)q_{2L-1}(x_*, -c) > 0. \quad (4.37)$$

At  $x = 0$ , the second-smallest singular value of  $A_{2L-1}$  is

$$s_{L,2}^{(0)} = \sqrt{1 + a^2 - 2a \cos \frac{\pi}{L}}.$$

The same appendix proves the strict separation

$$x_* + c < 2\sqrt{a} \sin \frac{\pi}{2L} \leq s_{L,2}^{(0)}. \quad (4.38)$$

The product parity and this second-singular-value separation now identify the relevant signed root independently of any prior branch choice.

Let  $\mu_1, \dots, \mu_{2L-1}$  be the eigenvalues of the real symmetric matrix

$$B_{2L-1}(x_*) = (x_* I_{2L-1} - A_{2L-1}) J_{2L-1}.$$

Right multiplication by  $J_{2L-1}$  preserves singular values, so the numbers  $|\mu_i|$  are precisely the singular values of  $x_* I_{2L-1} - A_{2L-1}$ . Moreover,

$$q_{2L-1}(x_*, c) q_{2L-1}(x_*, -c) = \prod_{i=1}^{2L-1} (\mu_i^2 - c^2).$$

Positivity of the product in (4.37) implies that the number of indices with  $|\mu_i| < c$  is even. On the other hand, Weyl's singular-value perturbation inequality and (4.38) give, for the second-smallest singular value  $s_{\text{next}}$ ,

$$s_{\text{next}}(x_* I_{2L-1} - A_{2L-1}) \geq s_{L,2}^{(0)} - x_* > c,$$

so at most one index can satisfy  $|\mu_i| < c$ . Parity therefore gives  $|\mu_i| \geq c$  for every  $i$ , and strict positivity of the product rules out equality. Hence  $s_{\min}(x_* I_{2L-1} - A_{2L-1}) > c_L$ . Since  $0 < x_* < 2r \sin \frac{\pi}{2L}$ , the point lies strictly in the positive central gap, and  $d_{L-1} > c_L$ . This completes the proof of (4.27). Together with the upper comparison, this proves the strict alternating chain  $c_m > d_m > c_{m+1}$ , exactly resolving the central-gap obstruction left by the noncentral chord comparison.  $\square$

*Proof of Theorem 4.1.* Reflection and the central interlacing chain now assemble the local comparisons into the global barrier inequality. Put  $N = n + 1$  and  $M = N + 1$ , so the larger and smaller matrices have orders  $M - 1$  and  $N - 1$ , respectively. If  $\Sigma = \text{diag}(1, -1, 1, -1, \dots)$ , then  $\Sigma A_n \Sigma = -A_n$ , so reflection reduces every noncentral comparison to a positive gap. For each positive noncentral gap  $G_{M-1,j}$ , choose  $x_j \in G_{M-1,j}^\circ$  attaining  $\gamma_{M-1,j}$ . Then  $j \geq 1$  and  $(j + 1)\pi/N \leq \pi/2$ , so theorem 4.3 produces a point  $\hat{x}_j \in G_{N-1,j}^\circ$ , and chord exhaustion gives

$$\gamma_{N-1,j} \geq g_{N-1}(\hat{x}_j) > g_{M-1}(x_j) = \gamma_{M-1,j}.$$

Reflection gives the same inequality on the corresponding reflected gap.

By theorem 4.4, if  $M = 2\ell_* + 1$ , the remaining single central height of the larger, even-order matrix is  $c_{\ell_*}$ , whereas each central height of the smaller odd-order matrix is  $d_{\ell_*-1} > c_{\ell_*}$ . If  $M = 2\ell_* + 2$ , the two central heights of the larger odd-order matrix equal  $d_{\ell_*} < c_{\ell_*}$ , where  $c_{\ell_*}$  is the single central height of the smaller even-order matrix. Thus every height of the larger problem is strictly smaller than at least one height of the smaller problem. Choose  $j_*$  with  $\gamma_{M-1, j_*} = \max_j \gamma_{M-1, j}$ . Then

$$\gamma_{M-1} = \gamma_{M-1, j_*} < \max_k \gamma_{N-1, k} = \gamma_{N-1}.$$

Since  $M - 1 = n + 1$  and  $N - 1 = n$ , this proves  $\gamma_{n+1} < \gamma_n$ .  $\square$

## 5. Quantitative bounds and the transition scale

The connectedness criterion (3.2) and the strict comparison of theorem 4.1 reduce the remaining problem to estimating  $\gamma_n$ . Put  $N = n + 1$  and recall  $\mathcal{L}_n(a)$  and  $\mathcal{U}_n(a)$  from (2.2)–(2.3).

**Proposition 5.1** (Sharp-order gap bounds). *For every  $n \geq 2$ ,*

$$\mathcal{L}_n(a) \leq \gamma_n \leq \mathcal{U}_n(a). \quad (5.1)$$

*In particular, both bounds have order  $(n + 1)a^{n/2}$ , with constants depending only on  $a$ .*

*Proof.* For  $x = 2r \cos \theta \in I_n^{\text{sp}}$ , set

$$\mathbf{v} = (v_j)_{j=1}^n, \quad v_j = r^{j-1} \sin(j\theta), \quad 1 \leq j \leq n.$$

The residual  $(xI_n - A_n)\mathbf{v}$  is supported in its last coordinate, where it equals  $r^n \sin(N\theta)$ . Thus

$$g_n(2r \cos \theta) \leq \frac{r^n |\sin(N\theta)|}{\left(\sum_{j=1}^n r^{2j-2} \sin^2(j\theta)\right)^{1/2}} \leq r^n \frac{|\sin(N\theta)|}{\sin \theta}.$$

Since  $\pi/N \leq \theta \leq (N - 1)\pi/N$ , this proves the upper bound.

For the lower bound, put  $p_\theta = re^{i\theta}$ ,  $q_\theta = re^{-i\theta}$ , so that  $p_\theta + q_\theta = x$  and  $p_\theta q_\theta = a$ . For  $\mathbf{u} = (u_j)_{j=1}^n \in \mathbb{C}^n$ , put  $u_0 = u_{n+1} = 0$ , and define maps from  $\mathbb{C}^n$  to  $\mathbb{C}^{n+1}$  by

$$(\mathbf{C}_\theta \mathbf{u})_j = p_\theta u_j - u_{j+1}, \quad (\mathbf{D}_\theta \mathbf{u})_j = u_j - \overline{q_\theta} u_{j+1}, \quad 0 \leq j \leq n.$$

Direct multiplication, using  $p_\theta + q_\theta = x$  and  $p_\theta q_\theta = a$ , gives

$$xI_n - A_n = D_\theta^* C_\theta. \quad (5.2)$$

The two Gram matrices are diagonally unitarily similar to the tridiagonal Toeplitz matrix with diagonal  $1 + r^2$  and off-diagonal  $-r$ . Its eigenvalues are  $1 + r^2 - 2r \cos \frac{k\pi}{N}$ ,  $1 \leq k \leq n$ . Consequently, for every  $\mathbf{u} \in \mathbb{C}^n$ ,

$$\begin{aligned} \|C_\theta \mathbf{u}\|_2 &\geq \left(1 + r^2 - 2r \cos \frac{\pi}{N}\right)^{1/2} \|\mathbf{u}\|_2, \\ \|D_\theta \mathbf{u}\|_2 &\geq \left(1 + r^2 - 2r \cos \frac{\pi}{N}\right)^{1/2} \|\mathbf{u}\|_2. \end{aligned} \quad (5.3)$$

Put

$$\mathcal{S}_N = \sum_{j=0}^n r^{2j} = \frac{1 - r^{2N}}{1 - r^2}.$$

Unit normal vectors to the two hyperplane ranges are

$$\mathbf{u}_C = \mathcal{S}_N^{-1/2}(\overline{p_\theta}^n, \overline{p_\theta}^{n-1}, \dots, 1)^T, \quad \mathbf{u}_D = \mathcal{S}_N^{-1/2}(1, q_\theta, \dots, q_\theta^n)^T.$$

In the polar decompositions

$$C_\theta = Q_C(C_\theta^* C_\theta)^{1/2}, \quad D_\theta = Q_D(D_\theta^* D_\theta)^{1/2},$$

the matrices  $Q_C, Q_D$  are isometries onto these ranges. Two hyperplanes in  $\mathbb{C}^{n+1}$  intersect in dimension  $n - 1$ , so the cross-Gram matrix  $Q_D^* Q_C$  has  $n - 1$  singular values equal to 1; its final singular value is  $|\langle \mathbf{u}_D, \mathbf{u}_C \rangle|$ , the cosine of the nontrivial principal angle. The geometric sum gives

$$\begin{aligned} |\langle \mathbf{u}_D, \mathbf{u}_C \rangle| &= \frac{r^n}{\mathcal{S}_N} \left| \sum_{j=0}^{N-1} e^{i(2j-(N-1))\theta} \right| \\ &= \frac{r^n(1 - r^2)}{1 - r^{2N}} \left| \frac{\sin(N\theta)}{\sin \theta} \right| =: \chi_N(\theta). \end{aligned} \quad (5.4)$$

Finally,

$$xI_n - A_n = (D_\theta^* D_\theta)^{1/2} Q_D^* Q_C (C_\theta^* C_\theta)^{1/2}.$$

Using  $s_{\min}(XYZ) \geq s_{\min}(X)s_{\min}(Y)s_{\min}(Z)$  and (5.3) gives

$$g_n(2r \cos \theta) \geq \left(1 + r^2 - 2r \cos \frac{\pi}{N}\right) \chi_N(\theta).$$

Putting  $\theta = 3\pi/(2N)$ , which lies in the first spectral gap, proves the lower bound.

Finally,

$$\mathcal{U}_n(a) \leq \frac{1}{2}(n+1)r^n,$$

while

$$\mathcal{L}_n(a) \geq C_{\text{lb}}(a)(n+1)r^n, \quad C_{\text{lb}}(a) = \frac{2(1-r)^2(1-r^2)}{3\pi}. \quad (5.5)$$

These follow from  $\sin \frac{\pi}{N} \geq 2/N$ ,  $1 + r^2 - 2r \cos \frac{\pi}{N} \geq (1-r)^2$ , and  $\sin \frac{3\pi}{2N} \leq 3\pi/(2N)$ . This proves the asserted sharp order.  $\square$

*Proof of Theorem 2.1.* By Theorem 5.1,  $\gamma_n \leq \mathcal{U}_n(a) \rightarrow 0$ . Theorem 4.1 says that  $\gamma_n$  is strictly decreasing. Equation (3.2) therefore shows that, once the pseudospectrum is connected, it stays connected in every larger dimension, which proves the set identity in part 3. Contractibility of every component was proved in section 3.1.

If  $\mathcal{L}_n(a) \geq \varepsilon$ , then  $\gamma_n \geq \varepsilon$ , so the pseudospectrum in dimension  $n$  is disconnected. If  $\mathcal{U}_n(a) < \varepsilon$ , then  $\gamma_n < \varepsilon$ , so it is connected. Strict decrease of  $\gamma_n$  now gives (2.4).

For  $t > 0$ , define

$$\Theta_n = (n+1)r^n, \quad \mathcal{T}_r(t) = \min\{N_0 \geq 2 : \Theta_m < t \text{ for every } m \geq N_0\}.$$

This minimum exists because  $\Theta_m \rightarrow 0$ . Because persistence gives

$$N_c(a, \varepsilon) = \min\{N \geq 2 : \gamma_m < \varepsilon \text{ for every } m \geq N\},$$

the bounds  $C_{\text{lb}}(a)\Theta_m \leq \gamma_m \leq \Theta_m/2$  immediately give

$$\mathcal{T}_r(\varepsilon/C_{\text{lb}}(a)) \leq N_c(a, \varepsilon) \leq \mathcal{T}_r(2\varepsilon). \quad (5.6)$$

Thus the polynomial factor  $n+1$ , as well as the exponential factor  $a^{n/2}$ , is sharp on both sides.

*Elementary inversion of the envelope threshold*

Let

$$\beta = -\log r = \frac{1}{2} \log(1/a), \quad L(t) = \log(1/t), \quad S_r(t) = \frac{L(t) + \log L(t)}{\beta}.$$

For every fixed  $D \in \mathbb{R}$ , direct substitution gives

$$\beta(S_r(t) + D) - \log(S_r(t) + D + 1) - L(t) = \beta D + \log \beta + o(1) \quad (t \downarrow 0).$$

Indeed,  $\beta S_r = L + \log L$  and  $\log(S_r + D + 1) = \log L - \log \beta + o(1)$ . Choose constants  $D_- < -(\log \beta)/\beta < D_+$ . For all sufficiently small  $t$ , the preceding

residual is negative at  $S_r(t) + D_-$  and positive at  $S_r(t) + D_+$ . Since  $S_r(t) \rightarrow \infty$  and

$$\frac{d}{dx}((x+1)e^{-\beta x}) = e^{-\beta x}(1 - \beta(x+1)),$$

this function is decreasing on  $[S_r(t) + D_-, \infty)$  for all sufficiently small  $t$ . The strict integer tail condition therefore gives

$$\lfloor S_r(t) + D_- \rfloor < \mathcal{T}_r(t) \leq \lceil S_r(t) + D_+ \rceil.$$

Thus  $\mathcal{T}_r(t) = S_r(t) + O_r(1)$ . Replacing  $t$  by a fixed positive multiple changes  $S_r(t)$  by only  $O_r(1)$ ; applying this to the two sides of (5.6) proves (2.5).

*Exact two-site endpoint*

For  $|x| \leq r$ ,

$$\det((xI_2 - A_2)^*(xI_2 - A_2) - a^2I_2) = x^2(x^2 - 2a - 2a^2) \leq 0.$$

The matrix inside the determinant is Hermitian, and the nonpositive product of its two eigenvalues forces its least eigenvalue to be nonpositive. Thus  $s_{\min}(xI_2 - A_2) \leq a$ . At  $x = 0$ , the singular values of  $-A_2$  are 1 and  $a$ , so equality is attained and  $\gamma_2 = a$ . Hence (3.2) gives  $N_c(a, \varepsilon) = 2$  if and only if  $\varepsilon > a$ . This proves all assertions.  $\square$

## 6. Numerical illustrations

Figures 1 and 2 illustrate two aspects of the matrix-order transition: coalescence of pseudospectral components and decay of the real-axis backward-error barrier.

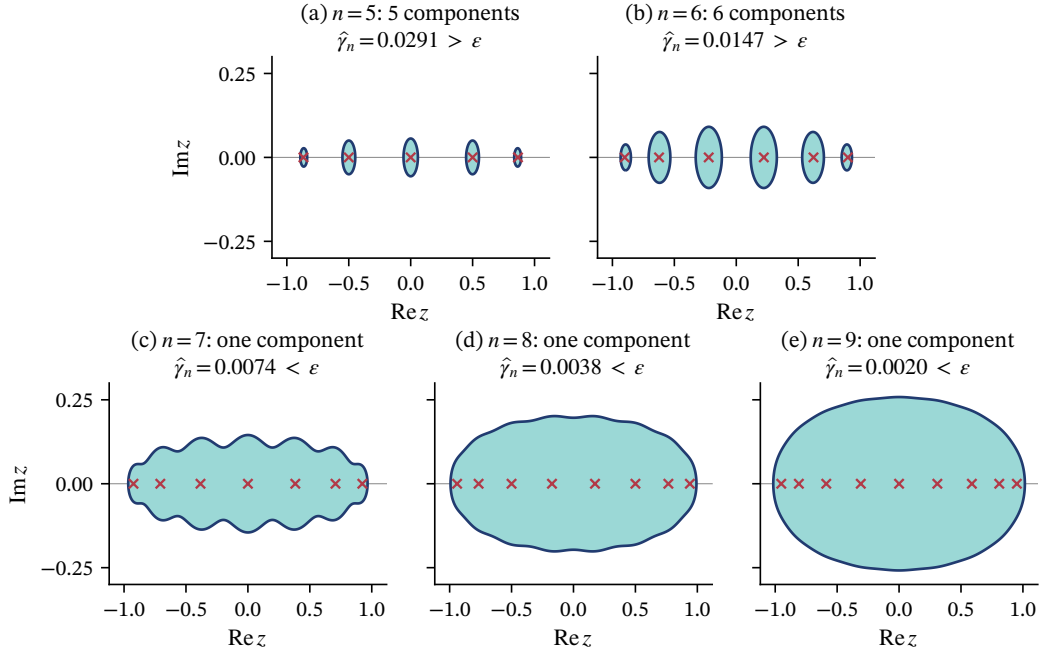


Figure 1: Numerically sampled strict pseudospectra of  $A_n(1/4)$  at  $\varepsilon = 10^{-2}$ . Shading marks the sampled pseudospectrum and crosses mark the exact eigenvalues. The samples show component coalescence between  $n = 6$  and  $n = 7$ .

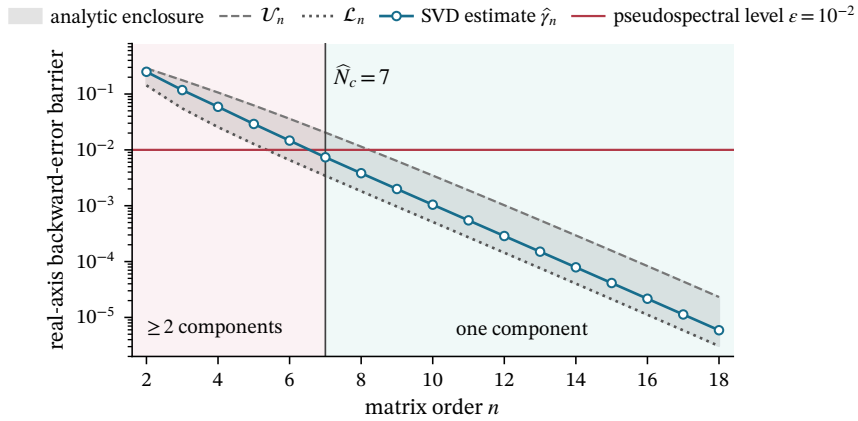


Figure 2: Real-axis backward-error barriers for  $a = 1/4$ . The grey band, bounded by the dotted lower curve  $\mathcal{L}_n(a)$  and dashed upper curve  $\mathcal{U}_n(a)$ , is the rigorous enclosure of  $\gamma_n$  and proves the scale  $(n+1)a^{n/2}$ . Blue circles joined by a solid line are floating-point estimates; the red horizontal line is the fixed pseudospectral level  $\varepsilon = 10^{-2}$ , and the black vertical line marks the numerical crossing  $\hat{N}_c = 7$ . The theorem gives the analytic bracket  $6 \leq N_c \leq 9$ .

## 7. Discussion

The principal conclusion is a finite-size order law for a global pseudospectral quantity. The sharp bounds identify the scale  $(n + 1)a^{n/2}$ , while strict dimension monotonicity ensures that connectedness, once attained, persists at every larger order.

The range  $0 < a < 1$  lies between two distinct endpoint regimes. At  $a = 0$ , the matrix is the nilpotent Jordan block  $V_n$ , and the diagonal symmetrizing similarity is singular. At  $a = 1$ , the matrix is normal; component coalescence is then governed by algebraic eigenvalue spacing rather than the exponential scale in theorem 5.1. The remainder  $O_a(1)$  in (2.5) is understood pointwise for fixed  $a$ ; the normal limit  $a \uparrow 1$  has a different level-spacing scale.

The pseudospectrum throughout is generated by the full complex spectral-norm ball. Restricting perturbations to Toeplitz, tridiagonal, real, or entrywise patterns changes the attainable set and may change its connectedness threshold; see [7] for Toeplitz-structured pseudospectra. Extending the strict order comparison to such structured sets, to block Toeplitz matrices, or to periodic  $k$ -Toeplitz families provides natural directions for further study.

### 7.1. Hatano–Nelson interpretation

The nearest-neighbor Hatano–Nelson hopping law, restricted here to an open chain and up to an overall sign, gives  $H_n = t_L A_n(a)$ , where  $a = t_R/t_L$  and  $t_L > t_R > 0$  [8, 9]. Writing  $h = \frac{1}{2} \log(t_L/t_R)$ , the diagonal similarity in (1.3) gives a right-eigenvector envelope  $e^{-h(j-1)}$  and skin depth  $\ell_{\text{skin}} = h^{-1}$  [2]. In the original units,

$$\begin{aligned}\varepsilon_*(n) &= t_L \gamma_n(a) \asymp_a t_L (n + 1) e^{-hn}, \\ N_c(a, \delta) &= \ell_{\text{skin}} \left( \log(1/\delta) + \log \log(1/\delta) \right) + O_a(1),\end{aligned}$$

where  $\delta = \varepsilon_{\text{phys}}/t_L$ . Thus the localization scale also controls pseudospectral coalescence, with the additional sharp factor  $n + 1$ .

### 7.2. Interpretation of component coalescence

Pseudospectral components form robust coarse spectral clusters under the unstructured norm bound. The number  $N_c$  marks their final coalescence:  $n < N_c$  gives at least two components, whereas  $n \geq N_c$  gives one.

In summary, the family (1.2) admits an exact real-axis connectedness criterion, contractible components, a strictly decreasing barrier sequence, and sharp exponential–polynomial bounds. The affine reduction in theorem 2.2 makes these conclusions properties of the full positive-off-diagonal nonsymmetric tridiagonal Toeplitz class.

## Appendix A. Pentadiagonal continuant certificate for vertical monotonicity

**Proposition A.1** (Vertical determinant certificate). *Fix  $0 < a < 1$ ,  $x \in \mathbb{R}$ ,  $Y \geq 0$ , and  $t \in \mathbb{R}$ , and extend the matrix notation by  $A_1 = [0]$ . For  $k \geq 1$ , set*

$$\begin{aligned}\mathcal{P}_k(Y, t) &= ((x + i\sqrt{Y})I_k - A_k)^*((x + i\sqrt{Y})I_k - A_k) - t^2 I_k, \\ F_k(Y, t) &= \det \mathcal{P}_k(Y, t),\end{aligned}$$

and set  $F_0 = 1$ . For every  $n \geq 1$ ,

$$\mathcal{P}_n(Y, t) > 0 \implies \partial_Y F_n(Y, t) > 0.$$

*Proof.* At a fixed value of  $Y$ , write

$$d = x^2 + Y + 1 + a^2 - t^2, \quad b = -(1 + a)x + i(1 - a)\sqrt{Y}.$$

For  $n \geq 2$ , the matrix  $\mathcal{P}_n(Y, t)$  has diagonal

$$d - 1, d, \dots, d, d - a^2,$$

whereas for  $n = 1$  its sole entry is  $d - 1 - a^2$ . In all dimensions its first upper diagonal is  $b$ , its first lower diagonal is  $\bar{b}$ , and its second off-diagonals are  $a$ . Let  $T_k$  be the determinant of the pure pentadiagonal Toeplitz matrix with diagonal  $d$  and these two off-diagonals. Set  $T_{-1} = 0, T_0 = 1$ , and put

$$D_k = T_k - T_{k-1}, \quad R_k = T_k - a^2 T_{k-1}. \quad (\text{A.1})$$

Indeed, a determinant is affine in each diagonal entry. Replacing the first diagonal entry  $d$  by  $d - 1$  subtracts the cofactor  $T_{k-1}$ , whereas replacing the last one by  $d - a^2$  subtracts  $a^2 T_{k-1}$ . Thus, for  $k < n$ ,  $D_k$  and  $R_k$  are, respectively, the leading and trailing principal minors of  $\mathcal{P}_n(Y, t)$ .

The following continuant calculation isolates all dependence on  $Y$ . With  $T_k = 0$  for  $k < 0$ , a Laplace expansion over the last two rows, grouped according to the largest displaced index in the contributing permutation, gives for every  $k \geq 1$

$$\begin{aligned}T_k &= dT_{k-1} - (|b|^2 - a^2)T_{k-2} - (2a^2d - a(b^2 + \bar{b}^2))T_{k-3} \\ &\quad - a^2(|b|^2 - a^2)T_{k-4} + a^4dT_{k-5} - a^6T_{k-6} - a^2\mathbf{1}_{\{k=2\}}.\end{aligned} \quad (\text{A.2})$$

For example,  $T_1 = d$  and  $T_2 = d^2 - |b|^2$ ; these also verify the only inhomogeneous coefficient in (A.2). Multiplying the recurrence by  $z^k$  and summing gives

$$\sum_{k \geq 0} T_k z^k = \frac{1 - a^2 z^2}{Q(z, Y)}, \quad (\text{A.3})$$

where

$$Q(z, Y) = 1 - dz + (|b|^2 - a^2)z^2 + (2a^2d - a(b^2 + \bar{b}^2))z^3 \\ + a^2(|b|^2 - a^2)z^4 - a^4dz^5 + a^6z^6.$$

Consequently, with  $\mathcal{D}(z, Y) = \sum_{k \geq 0} D_k z^k$ ,

$$\mathcal{D}(z, Y) = \frac{(1-z)(1-a^2z^2)}{Q(z, Y)}, \quad \sum_{k \geq 0} F_k(Y, t)z^k = (1-a^2z)\mathcal{D}(z, Y). \quad (\text{A.4})$$

For the second identity, first change the initial diagonal, producing  $D_k$ , and then change the final diagonal; its cofactor is  $D_{k-1}$ . Thus  $F_k = D_k - a^2D_{k-1}$ , which proves the asserted generating function. Direct substitution of  $d$  and  $b$  gives

$$\partial_Y Q = -z(1-z)(1+az)^2(1-a^2z).$$

Differentiating (A.4) therefore yields

$$\sum_{k \geq 0} \partial_Y F_k(Y, t)z^k = \varphi_a(z)\mathcal{D}(z, Y)^2, \quad \varphi_a(z) = \frac{z(1+az)(1-a^2z)^2}{1-az}. \quad (\text{A.5})$$

Suppose that  $\mathcal{P}_n(Y, t) > 0$ . Then

$$D_k > 0, \quad R_k > 0 \quad (0 \leq k < n). \quad (\text{A.6})$$

Put  $u_k = \sum_{j=0}^k D_j D_{k-j}$ , with  $u_k = 0$  for  $k < 0$ . The coefficients of  $\varphi_a$ , beginning with the coefficient of  $z$ , are

$$1, \quad 2a(1-a), \quad a^2(a^2-4a+2), \quad 2a^{k-1}(1-a)^2 \quad (k \geq 4).$$

Thus

$$\partial_Y F_n = u_{n-1} + 2a(1-a)u_{n-2} + a^2(a^2-4a+2)u_{n-3} \\ + 2(1-a)^2 \sum_{k=4}^n a^{k-1}u_{n-k}. \quad (\text{A.7})$$

The third coefficient is the sole term that may be negative; the following cofactor inequality controls its contribution. Let  $j, k \geq 1$ ,  $j+k = n-1$ . Deleting index  $k+1$  from  $\mathcal{P}_n$  leaves blocks of sizes  $k$  and  $j$ , with the second off-diagonal  $a$  as their sole coupling. In block form, the deleted principal submatrix is

$$\begin{bmatrix} L_k & a\mathbf{e}_k\mathbf{e}_1^T \\ a\mathbf{e}_1\mathbf{e}_k^T & R_j^{\text{mat}} \end{bmatrix}, \quad \det L_k = D_k, \quad \det R_j^{\text{mat}} = R_j.$$

Expanding once in each of the two coupling entries shows that its determinant is

$$D_k R_j - a^2 D_{k-1} R_{j-1} > 0. \quad (\text{A.8})$$

Deleting index  $j + 1$  instead gives the same inequality with  $j, k$  interchanged. Define

$$\vartheta_k = \frac{D_k}{D_{k-1}}, \quad \varpi_k = \frac{R_k}{R_{k-1}}.$$

Since  $T_k = \sum_{\ell=0}^k D_\ell$ , (A.1) gives

$$R_k = D_k + (1 - a^2) \sum_{\ell=0}^{k-1} D_\ell, \quad R_k - R_{k-1} = D_k - a^2 D_{k-1}. \quad (\text{A.9})$$

Hence  $\vartheta_k \leq a^2$  implies  $\varpi_k \leq 1$ . On dividing (A.8) and its companion by positive factors, we obtain

$$\vartheta_k \varpi_j > a^2, \quad \vartheta_j \varpi_k > a^2. \quad (\text{A.10})$$

If both  $\vartheta_j$  and  $\vartheta_k$  were at most  $a^2$ , then both  $\varpi_j$  and  $\varpi_k$  would be at most 1, contradicting (A.10). Hence, after interchanging  $j, k$  if necessary,  $\vartheta_j > a^2$ . Moreover,

$$\vartheta_j R_{j-1} - a^2 R_j = (\vartheta_j - a^2)(R_{j-1} - a^2 D_{j-1}),$$

and the second factor equals  $(1 - a^2) \sum_{\ell=0}^{j-1} D_\ell > 0$ . Therefore

$$\vartheta_j > a^2 \implies \varpi_j < \vartheta_j / a^2.$$

The first inequality in (A.10) now gives  $a^2 < \vartheta_k \varpi_j < \vartheta_j \vartheta_k / a^2$ , and therefore

$$\vartheta_j \vartheta_k > a^4. \quad (\text{A.11})$$

Consequently, for  $n \geq 3$ , applying (A.11) to  $j = p + 1$ ,  $k = n - 2 - p$  gives

$$D_{p+1} D_{n-2-p} > a^4 D_p D_{n-3-p}, \quad 0 \leq p \leq n - 3.$$

Summing and then adding the two positive endpoint terms gives

$$u_{n-1} = 2D_{n-1} + \sum_{j=1}^{n-2} D_j D_{n-1-j} > a^4 u_{n-3}. \quad (\text{A.12})$$

If the third coefficient in (A.7) is negative, write it as  $-\mathcal{B}_{\text{neg}}$ , where  $\mathcal{B}_{\text{neg}} = a^2(4a - 2 - a^2)$ . Since

$$a^4 - \mathcal{B}_{\text{neg}} = 2a^2(1 - a)^2 > 0,$$

equations (A.7) and (A.12) prove (3.1). If that coefficient is nonnegative, the implication is immediate. For  $n = 1$ , one has  $\partial_Y F_1 = 1$ ; for  $n = 2$ , one has  $\partial_Y F_2 = 2D_1 + 2a(1 - a) > 0$ . Thus (3.1) holds in every dimension, which proves the proposition.  $\square$

## Appendix B. Folded-continuant transfer and boundary residuals

This appendix supplies the transfer and generating identities underlying (4.10)–(4.11) and records every boundary sign and parity convention in the signed-pencil argument. Pair the coordinates in the order  $(1, n), (2, n-1), \dots$ . In this order the symmetric matrix  $(xI_n - A_n)J_n - sI_n$  is block tridiagonal. Its interior diagonal and off-diagonal blocks are

$$D_{\text{int}} = \begin{bmatrix} -s & x \\ x & -s \end{bmatrix}, \quad E_{\text{off}} = \begin{bmatrix} 0 & -1 \\ -a & 0 \end{bmatrix},$$

with lower block  $E_{\text{off}}^T$ . For even order the terminal block is

$$D_{\text{term}}^{\text{e}} = \begin{bmatrix} -1-s & x \\ x & -a-s \end{bmatrix};$$

for odd order the terminal block is the scalar  $x-s$ , coupled to the preceding block by the column  $(-1, -a)^T$ . Since

$$(xI_n - A_n - sJ_n) = ((xI_n - A_n)J_n - sI_n)J_n,$$

the determinant of this block matrix, multiplied by  $\det J_n$ , is  $q_n(x, s)$ . Set  $\mathcal{Q}_m^{\text{e}} = q_{2m}(x, s)$  and  $\mathcal{Q}_m^{\text{o}} = q_{2m+1}(x, s)$ . The following five-minor transfer completes the continuant verification. For  $\tau \in \{\text{e}, \text{o}\}$ , let  $K_m^\tau$  denote the corresponding matrix

$$(xI_n - A_n - sJ_n)J_n = (xI_n - A_n)J_n - sI_n, \quad n = 2m \text{ or } 2m+1,$$

after simultaneously permuting its rows and columns into the paired order. Thus  $K_m^{\text{e}}$  has order  $2m$ ,  $K_m^{\text{o}}$  has order  $2m+1$ , and

$$K_1^{\text{e}} = \begin{bmatrix} -1-s & x \\ x & -a-s \end{bmatrix}, \quad K_1^{\text{o}} = \begin{bmatrix} -s & x & -1 \\ x & -s & -a \\ -1 & -a & x-s \end{bmatrix}.$$

For  $m \geq 1$ , increasing  $m$  by one adds one interior block at the exposed end:

$$K_{m+1}^\tau = \begin{bmatrix} D_{\text{int}} & \mathcal{E}_m^\tau \\ (\mathcal{E}_m^\tau)^T & K_m^\tau \end{bmatrix}, \quad \mathcal{E}_m^\tau = \begin{bmatrix} E_{\text{off}} & 0 \end{bmatrix}, \quad (\text{B.1})$$

where the zero block has the width required by  $K_m^\tau$ .

For a matrix whose first two coordinates are exposed, write  $K(\widehat{I} \mid \widehat{J})$  for the matrix obtained by deleting rows  $I$  and columns  $J$ . Suppressing the parity superscript, set

$$\begin{aligned} \mathfrak{d}_m &= \det K_m, & \mathfrak{p}_m &= \det K_m(\widehat{1} \mid \widehat{1}), & \mathfrak{q}_m &= \det K_m(\widehat{2} \mid \widehat{2}), \\ \mathfrak{c}_m &= \det K_m(\widehat{1} \mid \widehat{2}), & \mathfrak{r}_m &= \det K_m(\widehat{1}, \widehat{2} \mid \widehat{1}, \widehat{2}). \end{aligned}$$

The off-principal minor obtained by deleting row 2 and column 1 has the same value  $\mathbf{c}_m$ , because  $K_m$  is symmetric.

The one-step transfer follows from a Schur complement. On the dense set where both  $K_m$  and the tail obtained by deleting its first two coordinates are invertible, let

$$F = \begin{bmatrix} u & w \\ w & v \end{bmatrix}$$

be its Schur complement in  $K_m$ , and let  $\delta$  be the determinant of that tail. Then

$$(\mathfrak{d}_m, \mathfrak{p}_m, \mathfrak{q}_m, \mathbf{c}_m, \mathfrak{r}_m) = \delta(uv - w^2, v, u, w, 1).$$

Since

$$E_{\text{off}} \text{adj}(F) E_{\text{off}}^T = \begin{bmatrix} u & -aw \\ -aw & a^2v \end{bmatrix},$$

taking the Schur complement of  $K_m$  in (B.1) gives

$$\begin{aligned} \mathfrak{d}_{m+1} &= (s^2 - x^2)\mathfrak{d}_m + s(a^2\mathfrak{p}_m + \mathfrak{q}_m) - 2ax\mathbf{c}_m + a^2\mathfrak{r}_m, \\ \mathfrak{p}_{m+1} &= -s\mathfrak{d}_m - a^2\mathfrak{p}_m, \\ \mathfrak{q}_{m+1} &= -s\mathfrak{d}_m - \mathfrak{q}_m, \\ \mathbf{c}_{m+1} &= x\mathfrak{d}_m + a\mathbf{c}_m, \\ \mathfrak{r}_{m+1} &= \mathfrak{d}_m. \end{aligned} \tag{B.2}$$

Both sides are polynomial in  $x$  and  $s$ . For each fixed  $0 < a < 1$ , the identities hold on a dense nonsingular set, so the polynomial identity theorem extends them to all  $x, s$ , including singular matrix cases.

The five quantities in (B.2) satisfy one linear invariant. Define

$$\ell_m = (1 - a)\mathfrak{d}_m - as\mathfrak{p}_m + s\mathfrak{q}_m + (1 - a)x\mathbf{c}_m - a(1 - a)\mathfrak{r}_m.$$

Direct substitution into (B.2) gives

$$\ell_{m+1} = -a\ell_m.$$

For the two terminal blocks, direct evaluation gives

$$\begin{aligned} &(\mathfrak{d}_1, \mathfrak{p}_1, \mathfrak{q}_1, \mathbf{c}_1, \mathfrak{r}_1)_e \\ &= ((1 + s)(a + s) - x^2, -a - s, -1 - s, x, 1), \\ &(\mathfrak{d}_1, \mathfrak{p}_1, \mathfrak{q}_1, \mathbf{c}_1, \mathfrak{r}_1)_o \\ &= (x(1 + a^2) + 2as - C_0(x - s), \\ &\quad s^2 - sx - a^2, s^2 - sx - 1, x^2 - sx - a, x - s). \end{aligned}$$

Thus  $\ell_1 = 0$  in both parity classes, and consequently

$$(1-a)\mathfrak{d}_m - as\mathfrak{p}_m + s\mathfrak{q}_m + (1-a)x\mathfrak{c}_m - a(1-a)\mathfrak{r}_m = 0 \quad (m \geq 1). \quad (\text{B.3})$$

Because

$$\det J_{2m} = \det J_{2m+1} = (-1)^m,$$

the first coordinate of the following signed state is the required determinant:

$$\boldsymbol{\eta}_m^\tau = (-1)^m \begin{bmatrix} \mathfrak{d}_m^\tau & \mathfrak{p}_m^\tau & \mathfrak{q}_m^\tau & \mathfrak{c}_m^\tau \end{bmatrix}^T, \quad (\boldsymbol{\eta}_m^e)_1 = \mathcal{Q}_m^e, \quad (\boldsymbol{\eta}_m^o)_1 = \mathcal{Q}_m^o.$$

Eliminating  $\mathfrak{r}_m$  with (B.3) reduces (B.2) to the four-dimensional transfer

$$\boldsymbol{\eta}_{m+1}^\tau = \mathsf{T}_{\text{fold}} \boldsymbol{\eta}_m^\tau, \quad \mathsf{T}_{\text{fold}} = \begin{bmatrix} x^2 - s^2 - a & \frac{a^3 s}{1-a} & -\frac{s}{1-a} & ax \\ s & a^2 & 0 & 0 \\ s & 0 & 1 & 0 \\ -x & 0 & 0 & -a \end{bmatrix}. \quad (\text{B.4})$$

Set

$$\boldsymbol{\eta}_0^e = \begin{bmatrix} 1 & a^{-1} & 1 & 0 \end{bmatrix}^T, \quad \boldsymbol{\eta}_0^o = \begin{bmatrix} x-s & 1 & 1 & -1 \end{bmatrix}^T. \quad (\text{B.5})$$

These values incorporate the terminal conditions at  $m = 0$ . Direct multiplication in (B.4) reproduces the signed terminal minors above. Hence  $\boldsymbol{\eta}_m^\tau = \mathsf{T}_{\text{fold}}^m \boldsymbol{\eta}_0^\tau$  for every  $m \geq 0$ .

A direct determinant calculation gives

$$\begin{aligned} \det(\lambda I_4 - \mathsf{T}_{\text{fold}}) &= \lambda^4 - C_0 \lambda^3 + \mathfrak{b} \lambda^2 - a^2 C_0 \lambda + a^4, \\ \mathfrak{b} &= (1+a^2)x^2 + 2as^2 + 2a^2 - 2a - 2a^3. \end{aligned} \quad (\text{B.6})$$

Apply Cayley–Hamilton to  $\mathsf{T}_{\text{fold}}$ , multiply by  $\mathsf{T}_{\text{fold}}^{m-4} \boldsymbol{\eta}_0^\tau$ , and extract the first coordinate. For both  $Z_m = \mathcal{Q}_m^e$  and  $Z_m = \mathcal{Q}_m^o$ , this yields

$$Z_m - C_0 Z_{m-1} + \mathfrak{b} Z_{m-2} - a^2 C_0 Z_{m-3} + a^4 Z_{m-4} = 0, \quad m \geq 4. \quad (\text{B.7})$$

Finally, applying  $\mathsf{T}_{\text{fold}}$  three times to (B.5) gives the boundary residuals

$$\begin{aligned} \text{Res}_0 &= Z_0, & \text{Res}_1 &= Z_1 - C_0 Z_0, & \text{Res}_2 &= Z_2 - C_0 Z_1 + \mathfrak{b} Z_0, \\ \text{Res}_3 &= Z_3 - C_0 Z_2 + \mathfrak{b} Z_1 - a^2 C_0 Z_0, \end{aligned}$$

namely

|                         | Res <sub>0</sub> | Res <sub>1</sub>      | Res <sub>2</sub>    | Res <sub>3</sub> |
|-------------------------|------------------|-----------------------|---------------------|------------------|
| $Z_m = \mathcal{Q}_m^e$ | 1                | $a(1 - 2\zeta_s)$     | $a^2(1 - 2\zeta_s)$ | $a^3$            |
| $Z_m = \mathcal{Q}_m^o$ | $x - s$          | $-[x(1 + a^2) + 2as]$ | $a^2(x - s)$        | 0                |

Thus, as a formal-power-series identity,

$$\Pi(z) \sum_{m \geq 0} Z_m z^m = \text{Res}_0 + \text{Res}_1 z + \text{Res}_2 z^2 + \text{Res}_3 z^3.$$

For the even sequence the polynomial on the right factors as  $(1+az)(1-2a\zeta_s z + a^2 z^2)$ ; for the odd sequence it is  $(x-s) - [x(1+a^2) + 2as]z + a^2(x-s)z^2$ . These factorizations identify the boundary-residual polynomials. Since  $\Pi(0) = 1$ , division by  $\Pi$  in  $\mathbb{R}[[z]]$  gives the two literal formal-power-series identities

$$\sum_{m \geq 0} \mathcal{Q}_m^e z^m = (1+az)(1-2a\zeta_s z + a^2 z^2) \Pi(z)^{-1}, \quad (\text{B.8})$$

and

$$\sum_{m \geq 0} \mathcal{Q}_m^o z^m = ((x-s) - [x(1+a^2) + 2as]z + a^2(x-s)z^2) \Pi(z)^{-1}. \quad (\text{B.9})$$

### Appendix C. Divided differences for the folded Chebyshev formulas

The preceding transfer calculation gives two rational generating functions. Coefficient extraction converts them to the divided-difference formulas used in the half-angle chord parametrization. Fix  $0 < a < 1$  and set  $U_{-1} = 0$ . The Chebyshev generating function gives

$$\sum_{m \geq 0} (u - \zeta_s)(U_m(u) + U_{m-1}(u)) \omega_{\text{gf}}^m = \frac{(u - \zeta_s)(1 + \omega_{\text{gf}})}{1 - 2u\omega_{\text{gf}} + \omega_{\text{gf}}^2},$$

and, with  $\mathbf{B}_s^{\text{fold}} = [x(1+a^2) + 2as]/(2a)$ ,

$$\sum_{m \geq 0} ((x-s)u - \mathbf{B}_s^{\text{fold}}) U_m(u) \omega_{\text{gf}}^m = \frac{(x-s)u - \mathbf{B}_s^{\text{fold}}}{1 - 2u\omega_{\text{gf}} + \omega_{\text{gf}}^2}.$$

Let  $u, v \in \mathbb{R}$  satisfy the two symmetric relations in (4.9). Taking the corresponding divided difference of the Chebyshev generating identities and putting  $\omega_{\text{gf}} = az$  recovers (B.8) and (B.9). Coefficient comparison gives (4.10)–(4.11), including the derivative value when  $u = v$ . The underlying polynomial identities remain valid when  $\Delta_{\text{fold}} < 0$ , without choosing a square-root branch.

### Appendix D. Exhaustion of the endpoint-side chord sheet

**Lemma D.1** (Endpoint-side chord-sheet exhaustion). *Let  $K \geq 3$  and let  $j \geq 1$  be an integer. Under the conventions of equations (4.15) and (4.16), if*

$$\frac{(j+1)\pi}{K} \leq \frac{\pi}{2},$$

then every point of the ordered middle branch in the corresponding open positive gap has a selected endpoint-side  $j$ -chord representation with  $j\pi/K < \theta < (j+1)\pi/K$ , and every  $\theta$  in this interval gives a point of that branch. The signed root is  $(-1)^j s$ , its magnitude is  $s_{\min}(xI_{K-1} - A_{K-1})$ , and the two folded variables are distinct. This includes a gap adjacent to zero. Separately, if  $K$  is odd and  $j = (K-1)/2$ , the same bidirectional statement holds from the positive spectral endpoint to the center of the single central gap, with  $j\pi/K < \theta \leq \pi/2$ .

*Proof.* Continue the ordered middle root  $s_*(x)$  through the open gap. Write  $\Delta_*(x) = \Delta_{\text{fold}}(x, s_*(x))$  for its folded discriminant, let  $y(x) \leq z(x)$  be the reconstructed nonnegative half variables on the real sheet issuing from the endpoint germ, and set  $\Theta(x) = \arccos y(x)$ . Let  $\mathcal{C}$  be the set of gap points for which

$$\Delta_*(x) > 0, \quad \frac{j\pi}{K} < \Theta(x) < \frac{(j+1)\pi}{K}, \quad u_K^{\max} < z(x) < \mathfrak{c}_h.$$

On  $\mathcal{C}$ , the reversible folded determinant identities give  $\mathcal{W}_K(y(x)) = \mathcal{W}_K(z(x))$ ; uniqueness on the endpoint-side outer lobe then gives

$$z(x) = z_K(\mathcal{W}_K(y(x))).$$

Thus  $\mathcal{C}$  is precisely the selected endpoint-side chord sheet. The endpoint germ constructed in the main text makes  $\mathcal{C}$  nonempty. The ordered eigenvalues of the continuous real symmetric pencil  $B_{K-1}(x)$  are continuous in  $x$ , including at an eigenvalue crossing, so the continuation passes through a multiple least singular value. The folded pair  $u = 2z^2 - 1$ ,  $v = 2y^2 - 1$  and its reconstructed half variables are continuous, while the sheet inequalities are strict. Thus  $\mathcal{C}$  is relatively open in the gap.

It is also relatively closed. At a relative closure point the chord identity and all sheet inequalities persist weakly. The possible closure cases are as follows. (1) If  $\Delta_*(x) = 0$ , then the reconstructed half variables coincide, which is impossible because

$$z \geq u_K^{\max} > \cos \frac{\pi}{K} \geq \cos \frac{j\pi}{K} \geq y.$$

(2) At an interior point of a positive gap, including one adjacent to zero, both half variables retain their signs:  $z > 0$ , while  $y = 0$  would force  $x = 0$ , outside the open positive half. (3) Neither the outer variable can reach  $\mathfrak{c}_h$  nor the inner variable a nodal endpoint: in either case the common level is zero, the endpoint-side solution is  $z = \mathfrak{c}_h$ , and (4.16) gives  $s = 0$ , impossible inside a

spectral gap. (4) The outer variable remains in its endpoint-side component, since its first exit point would be  $u_K^{\max}$ , contradicting (4.19). Passage through  $z = 1$  is harmless; only its coordinate changes. Hence every relative closure point remains in  $\mathcal{C}$ . Since the open gap is an interval,  $\mathcal{C}$  is the entire gap. This proves the first exhaustion direction: every actual gap point lies on the selected endpoint-side chord sheet.

For the converse direction, the reconstructed angle  $\Theta$  extends continuously to the closed spectral gap. The exact folded variables at the two endpoints give

$$\Theta(\lambda_{K-1,j+1}) = \frac{(j+1)\pi}{K}, \quad \Theta(\lambda_{K-1,j}) = \frac{j\pi}{K}.$$

For every  $\theta_0 \in (j\pi/K, (j+1)\pi/K)$ , the intermediate value theorem therefore supplies an interior gap point  $x$  with  $\Theta(x) = \theta_0$ . The classification just proved and uniqueness of the endpoint-side outer solution then give

$$y(x) = \cos \theta_0, \quad z(x) = z_K(\mathcal{W}_K(\cos \theta_0)),$$

so the corresponding abscissa and signed root are exactly those of the prescribed  $j$ -chord. Thus every chord parameter produces an actual point of the ordered middle branch. By theorem 4.2, its positive magnitude is  $s_{\min}(xI_{K-1} - A_{K-1})$ . Both exhaustion directions follow from continuity and sheet classification; monotonicity of  $\theta \mapsto x$  is unnecessary.

When  $K$  is odd, the same argument applies to the positive half of the single central gap, continuing from the positive spectral endpoint to  $x = 0$ . Because  $z > 0$ , the identity  $x = (2r/\mathfrak{c}_h) \cos \theta z$  forces  $\theta = \pi/2$  at  $x = 0$ . Thus every parameter on that half occurs, and reflection covers the negative half.  $\square$

## Appendix E. Scalar estimate for the lower central-gap comparison

This appendix proves the strict scalar inequality (4.35) through finite estimates valid uniformly over the three cases specified below.

Put  $b = u_+/u_- = 1/a$ , and define

$$Q_b = b^2 - 2b\rho + 1, \quad S_b = b^2 + 2b\omega_L + 1.$$

The Chebyshev identity gives  $u_-^2 Q_b = 1$ , and

$$\mathfrak{D}_L = S_b/Q_b, \quad \Delta_\xi := \rho - \xi = \Delta_\omega Q_b/S_b. \quad (\text{E.1})$$

Indeed,  $u_+ = bu_-$  and  $u_-^2 = Q_b^{-1}$ , so

$$\mathfrak{D}_L = 1 + 2(\rho + \omega_L)u_-u_+ = 1 + \frac{2b(\rho + \omega_L)}{Q_b} = \frac{S_b}{Q_b},$$

and the formula for  $\Delta_\xi$  follows from (4.34). The ratio  $U_L/U_{L-1}$  is strictly increasing above the largest zero of  $U_{L-1}$ , and equals 1 at  $\rho_{\text{ref}} = \cos \frac{\pi}{2L+1}$ . Hence  $\rho > \rho_{\text{ref}}$ . Since  $S_b - Q_b = 2b(\rho + \omega_L) > 0$ ,

$$\xi > \rho_{\text{ref}} - \Delta_\omega > 2\omega_L - 1 > 2\omega_L^2 - 1 = \cos \frac{\pi}{L}.$$

Therefore  $f_L(x) = (1+x)U_{L-1}(x)$  is positive, strictly increasing, and strictly log-concave on  $[\xi, \rho]$ . In particular,  $\mathfrak{d}_\rho = (\log f_L)'(\rho) > 0$ , and

$$\mathfrak{d}_\rho = \begin{cases} \frac{L(b-\rho)-1}{\rho^2-1}, & \rho \neq 1, \\ \frac{2L^2+1}{6}, & \rho = 1 \end{cases}.$$

The endpoint value follows directly from  $U_{L-1}(1) = L$  and  $U'_{L-1}(1) = L(L^2 - 1)/3$ . Moreover,

$$\log f_L(\rho) - \log f_L(\xi) > \Delta_\xi \mathfrak{d}_\rho, \quad f_L(\xi)^2 < f_L(\rho)^2 e^{-2\Delta_\xi \mathfrak{d}_\rho} < \frac{f_L(\rho)^2}{1 + 2\Delta_\xi \mathfrak{d}_\rho}.$$

Here the first strict inequality is the integral form of strict log-concavity. Since  $f_L(\rho)^2 = (1+\rho)^2/Q_b$  and  $\Delta_\xi = \Delta_\omega Q_b/S_b$ , substitution into (4.35) reduces it to

$$\mathcal{N} = (\rho + \omega_L)(S_b + 2\Delta_\omega Q_b \mathfrak{d}_\rho) - (1 + \omega_L)(1 + \rho)^2 > 0. \quad (\text{E.2})$$

All three cases use

$$\Delta_\omega = 1 - \cos \frac{\pi}{2L} < \frac{\pi^2}{8L^2} < \frac{5}{4L^2}, \quad \Delta_\omega L < 1. \quad (\text{E.3})$$

Suppose first that  $\rho > 1$ . Put  $u = b - \rho$  and  $\Delta_+ = \rho^2 - 1$ . Since  $\mathfrak{d}_\rho > 0$ ,  $u > 1/L$ . Keeping  $\rho$  fixed, direct differentiation gives

$$\frac{\partial_b \mathcal{N}}{2(\rho + \omega_L)} = \rho + \omega_L + u - \Delta_\omega L + \frac{\Delta_\omega(3Lu^2 - 2u)}{\Delta_+}.$$

The right-hand side is strictly increasing for  $u \geq 1/L$  and is positive at  $u = 1/L$  because  $\Delta_\omega L < 1$ . Hence  $\mathcal{N}$  exceeds its value at  $b = \rho + 1/L$ . With  $s_0 = \rho + \omega_L$  and  $\omega_+ = 1 + \omega_L$ , that value is

$$\Upsilon(s_0) = s_0 \left[ \left( s_0 + \frac{1}{L} \right)^2 + \omega_+ \Delta_\omega \right] - \omega_+ (s_0 + \Delta_\omega)^2.$$

For  $s_0 \geq \omega_+$ ,

$$\Upsilon''(s_0) = 6s_0 + 4/L - 2\omega_+ > 0, \quad \Upsilon'(\omega_+) = 2\omega_L \omega_+ + 4\omega_+/L + 1/L^2 > 0,$$

and

$$\Upsilon(\omega_+) = \omega_+(-2\Delta_\omega + 2\omega_+/L + 1/L^2) > 0.$$

Thus (E.2) holds when  $\rho > 1$ . If  $\rho = 1$ , then  $b = (L + 1)/L$ ,  $Q_b = L^{-2}$ , and direct substitution, using the endpoint value of  $\mathfrak{d}_\rho$ , gives

$$\mathcal{N} = \Upsilon(1 + \omega_L) + \frac{2(1 + \omega_L)\Delta_\omega}{L^2} \frac{2L^2 + 1}{6} > 0.$$

The first term is positive by the preceding calculation, and every factor in the correction term is positive.

For  $\rho < 1$ , set

$$v = 1 - \rho, \quad \Delta_b = b - 1, \quad u = \Delta_b + v, \quad \Delta_- = v(2 - v).$$

Because  $b > 1$ , we have  $0 < v < u$ . Moreover,  $\rho^2 - 1 < 0$ , so the positivity of  $\mathfrak{d}_\rho$  in its quotient formula forces  $L(b - \rho) - 1 < 0$ . Since  $u = b - \rho$ ,

$$0 < v < u < 1/L. \quad (\text{E.4})$$

With  $\rho$  fixed, write  $\mathcal{N}_\rho^<(\cdot)$  for  $\mathcal{N}$  viewed as a function of  $\Delta_b = b - 1$ . A direct expansion of (E.2) gives

$$\mathcal{N}_\rho^<(\Delta_b) - \mathcal{N}_\rho^<(0) = \frac{\Delta_b(\rho + \omega_L)}{\Delta_-} \mathfrak{q}(u),$$

where

$$\mathfrak{q}(u) = \Delta_-[2(2 - \Delta_\omega) + u - v] + 2\Delta_\omega[u + v - L(u^2 + uv + v^2) - L\Delta_-].$$

The function  $\mathfrak{q}$  is a concave quadratic. Moreover,

$$\mathcal{N}_\rho^<(0) = \frac{v}{2 - v} \left[ (2 - \Delta_\omega)(2 - v)^2 - 2\Delta_\omega^2 - 4\Delta_\omega L(2 - v - \Delta_\omega) \right].$$

The following elementary bounds hold; the second follows from  $\rho > \rho_{\text{ref}}$  and  $1 - \rho_{\text{ref}} < \pi^2/[2(2L + 1)^2] < 5/(2L + 1)^2$ :

$$\Delta_\omega < \frac{5}{4L^2}, \quad v < \frac{5}{(2L + 1)^2} \leq \frac{1}{5}. \quad (\text{E.5})$$

Although  $5/(2L + 1)^2 = 1/5$  when  $L = 2$ , the preceding strict bound still gives  $v < 1/5$ . These bounds show that the bracketed expression is positive, since it equals

$$(2 - v) \left[ (2 - \Delta_\omega)(2 - v) - 4\Delta_\omega L \right] + 2\Delta_\omega^2(2L - 1)$$

and, using  $L \geq 2$ ,

$$(2 - \Delta_\omega)(2 - v) > \frac{27}{16} \frac{9}{5} = \frac{243}{80} > \frac{5}{2} \geq 4\Delta_\omega L.$$

At the endpoints of (E.4),

$$\mathfrak{q}(v) = v[(2 - v)(4 - 2\Delta_\omega) + 4\Delta_\omega - 4\Delta_\omega L(1 + v)] > 0,$$

$$\mathfrak{q}(1/L) = v[(2 - v)(4 - 2\Delta_\omega + 1/L - v) - 4\Delta_\omega L] > 0.$$

At the two endpoints, the positive leading terms exceed

$$\frac{9}{5} \left(4 - \frac{5}{8}\right) = \frac{243}{40}, \quad \frac{9}{5} \left(4 - \frac{5}{8} - \frac{1}{5}\right) = \frac{1143}{200},$$

while the negative terms have magnitudes less than  $(5/L)(6/5) \leq 3$  and  $5/L \leq 5/2$ , respectively. Concavity therefore gives  $\mathfrak{q}(u) > 0$ , proving (E.2), and hence (4.35), in all three cases.

## Appendix F. Signed-root certification in the lower central gap

This appendix verifies the folded substitution and second-singular-value separation used in the main proof of  $d_m > c_{m+1}$ .

At  $x = x_*$  and  $s = \pm c$ , the real pair  $u = \xi$ ,  $v = -\omega_L$  satisfies the two symmetric relations in (4.9). Indeed, its sum and product are  $\xi - \omega_L$  and  $-\omega_L \xi$ , and the required identities follow from

$$\mathfrak{D}_L = \frac{\zeta + \omega_L}{\zeta - \rho}, \quad \xi = \frac{\rho(\zeta + 1) + \zeta(\omega_L - 1)}{\zeta + \omega_L}.$$

Positivity of  $\xi + \omega_L$  selects the distinct-variable branch of the divided difference in (4.11). For convenience, write the folded numerator as

$$\mathfrak{n}_s(t) = -[x(\zeta - t) + s(1 + t)]U_{L-1}(t).$$

Put

$$\mathbf{X}_\xi = x_*(\zeta - \xi), \quad \mathbf{Y}_\xi = c(1 + \xi), \quad \mathbf{X}_\omega = x_*(\zeta + \omega_L), \quad \mathbf{Y}_\omega = c\Delta_\omega,$$

and set  $U_\xi = U_{L-1}(\xi)$ ,  $U_{-\omega} = U_{L-1}(-\omega_L)$ . The identities

$$\mathfrak{D}_L = 2a(\zeta + \omega_L)u_+^2, \quad \mathfrak{D}_L(\rho + \omega_L)(\zeta - \xi) = (\zeta + \omega_L)(1 + \xi)$$

give  $Y_\omega Y_\xi = X_\omega X_\xi$ . The identity  $u_-^2 Q_b = 1$  gives  $Q_b > 0$ , and  $S_b - Q_b = 2b(\rho + \omega_L) > 0$ , so  $\mathfrak{D}_L = S_b/Q_b > 1$ . Moreover,  $L \geq 2$  gives  $\omega_L > 1/2$ , while  $\rho > \rho_{\text{ref}} > \omega_L$ ; hence

$$\rho + \omega_L > 2\omega_L > 1 > 1 - \omega_L > 0.$$

Consequently,

$$\frac{X_\omega^2}{Y_\omega^2} = \frac{\mathfrak{D}_L(\rho + \omega_L)}{1 - \omega_L} > 1,$$

so, with  $\mathfrak{r}_* = X_\xi/Y_\xi = Y_\omega/X_\omega \in (0, 1)$ , the common factor  $a^{L-1}/(\xi + \omega_L)$  in (4.11) is positive, and the two numerator differences are

$$\begin{aligned} \mathfrak{n}_c(\xi) - \mathfrak{n}_c(-\omega_L) &= (1 + \mathfrak{r}_*)(X_\omega U_{-\omega} - Y_\xi U_\xi), \\ \mathfrak{n}_{-c}(\xi) - \mathfrak{n}_{-c}(-\omega_L) &= (1 - \mathfrak{r}_*)(X_\omega U_{-\omega} + Y_\xi U_\xi). \end{aligned}$$

Their product is positive by (4.35), because

$$U_{-\omega}^2 = \frac{1}{1 - \omega_L^2}, \quad X_\omega^2 U_{-\omega}^2 = \frac{c^2 \mathfrak{D}_L(\rho + \omega_L)}{1 + \omega_L} > Y_\xi^2 U_\xi^2.$$

Thus

$$q_{2L-1}(x_*, c)q_{2L-1}(x_*, -c) > 0. \quad (\text{F.1})$$

At  $x = 0$ , the second-smallest singular value of  $A_{2L-1}$  is

$$s_{L,2}^{(0)} = \sqrt{1 + a^2 - 2a \cos \frac{\pi}{L}}.$$

Indeed, after separating odd and even coordinates, the two rectangular off-diagonal blocks have Gram matrices equal to the  $(L-1)$ -dimensional tridiagonal Toeplitz matrix with diagonal  $1 + a^2$  and off-diagonal  $a$ . Their singular values therefore have squares

$$1 + a^2 + 2a \cos \frac{k\pi}{L}, \quad 1 \leq k \leq L-1,$$

and each occurs twice, while the remaining singular value is zero. We claim

$$x_* + c < 2\sqrt{a} \sin \frac{\pi}{2L} \leq s_{L,2}^{(0)}. \quad (\text{F.2})$$

Indeed, recall  $\rho_{\text{ref}} = \cos \frac{\pi}{2L+1}$ . Since  $\rho > \rho_{\text{ref}}$  and  $U_{L-1}$  is strictly increasing above its largest zero,

$$u_- = U_{L-1}(\rho) > U_{L-1}(\rho_{\text{ref}}) =: u_{-, \text{ref}}, \quad u_{-, \text{ref}}^{-2} = 2(1 - \rho_{\text{ref}}).$$

The identities  $c = a/u_-$  and  $c^2 = 2a(\zeta - \rho)$  imply  $\zeta - \rho < a(1 - \rho_{\text{ref}}) < 1 - \rho_{\text{ref}}$ . Also

$$\frac{x_*^2}{2a} = \Delta_\omega \frac{\rho + \omega_L}{\zeta + \omega_L} < \Delta_\omega.$$

Put

$$\theta_L = \frac{\pi}{4L}, \quad \Delta_L = \frac{\theta_L}{2L+1}.$$

Then  $\Delta_\omega = 2 \sin^2 \theta_L$  and  $1 - \rho_{\text{ref}} = 2 \sin^2(\theta_L - \Delta_L)$ . Since all quantities are positive, the preceding quotient bound for  $x_*^2$  gives

$$x_* < 2\sqrt{a} \sin \theta_L.$$

Likewise,

$$c^2 = 2a(\zeta - \rho) < 2a^2(1 - \rho_{\text{ref}}) = 4a^2 \sin^2(\theta_L - \Delta_L),$$

and hence

$$c < 2a \sin(\theta_L - \Delta_L) < 2\sqrt{a} \sin(\theta_L - \Delta_L),$$

where the last inequality uses  $0 < a < 1$ , hence  $a < \sqrt{a}$ . It suffices to prove

$$\sin \theta_L + \sin(\theta_L - \Delta_L) < \sin(2\theta_L). \quad (\text{F.3})$$

The estimates

$$2 \sin \theta_L - \sin(2\theta_L) = 2 \sin \theta_L (1 - \cos \theta_L) < \theta_L^3$$

and

$$\sin \theta_L - \sin(\theta_L - \Delta_L) = \int_{\theta_L - \Delta_L}^{\theta_L} \cos u \, du > \Delta_L \cos \theta_L$$

reduce (F.3) to  $\Delta_L \cos \theta_L > \theta_L^3$ . Substituting  $\Delta_L = \theta_L/(2L+1)$  and  $\theta_L = \pi/(4L)$  gives the equivalent condition  $2 \cos \theta_L > \theta_L(\pi + 2\theta_L)$ . For  $0 < \theta_L \leq \pi/8$ , using  $\pi^2 < 10$ ,

$$2 \cos \theta_L > 2 - \theta_L^2 \geq \frac{59}{32} > \frac{25}{16} \geq \theta_L(\pi + 2\theta_L).$$

Consequently  $x_* + c < 2\sqrt{a} \sin \frac{\pi}{2L}$ . Finally,

$$(s_{L,2}^{(0)})^2 - 4a \sin^2 \frac{\pi}{2L} = (1-a)^2 \geq 0,$$

which proves (4.38).

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## Data availability

This theoretical study used no experimental data or external research datasets. Python code for the numerical illustrations and Lean 4 code accompanying the underlying mathematical results are available at <https://github.com/jinshanmu/Hatano-Nelson>.

## Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used OpenAI ChatGPT to assist with checking algebraic consistency and presentation. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

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